

Module-3

Basic Concepts of Disease and Therapy

Cellular Adaptations

Adaptations are reversible changes in the number, size, phenotype, metabolic activity, or functions of cells in response to changes in their environment. It can be physiologic or pathologic.

Physiologic adaptations usually represent responses of cells to normal stimulation by hormones or endogenous chemical mediators e.g., the hormone-induced enlargement of the breast and uterus during pregnancy. Pathologic adaptations are responses to stress that allow cells to modulate their structure and function and thus will be able to escape injury. Adaptations can take several distinct forms: hypertrophy, hyperplasia, atrophy or meta-plasia.

Hypertrophy: is an increase in the size of cells resulting in increase in the size of the organ. In pure hypertrophy there are no new cells, just bigger cells, enlarged by an increased amount of structural proteins and organelles.

Hypertrophy usually occurs when cells are incapable of dividing. But hypertrophy and hyperplasia can also occur together. Hypertrophy can be physiologic or pathologic.

Hypertrophy is caused either by increased functional demand or by specific hormonal stimulation. The massive physiologic enlargement of the uterus during pregnancy occurs as a consequence of estrogen stimulated smooth muscle hypertrophy and hyperplasia. The striated muscle cells in both the skeletal muscle and the heart can undergo only hypertrophy in response to increased demand because in the adult they have limited capacity to divide.

Examples of pathologic cellular hypertrophy include the cardiac enlargement that occurs with hypertension or aortic valve disease. The mechanisms driving cardiac hypertrophy involve at least two types of signals: mechanical triggers (such as stretch) and trophic triggers (such as activation of α -adrenergic receptors). These stimuli turn on signal transduction pathways that lead to the induction of a number of genes, which in turn stimulate synthesis of numerous cellular proteins, including growth factors and structural proteins. The result is the synthesis of more proteins and myo-filaments per cell, which achieves improved performance and thus a balance between the demand and the cell's functional capacity. A limit is reached beyond which the enlargement of muscle mass can no longer compensate for the increased burden. When this happens in the heart, several "degenerative" changes occur in the myocardial fibers. The net result of these changes is ventricular dilation and ultimately cardiac failure, a sequence of events that illustrates how an adaptation to stress can progress to functionally significant cell injury if the stress is not relieved.

Hyperplasia: is characterized by an increase in cell number. Takes place if the cell population is capable of replication. It may occur with hypertrophy and often in response to the same stimuli and can be physiologic or pathologic. There are two types of physiologic hyperplasia: hormonal hyperplasia and compensatory hyperplasia.

Hormonal hyperplasia: occurs when cells are stimulated by hormones, exemplified by the proliferation of the glandular epithelium of the female breast at puberty and during pregnancy.

Compensatory hyperplasia: occurs when a portion of the tissue is removed or diseased. For example when a liver is partially resected, mitotic activity in the remaining cells begins as early as 12 hours later, eventually restoring the liver to its normal weight. The stimuli are polypeptide growth factors produced by remnant hepatocytes as well as non-parenchymal cells in the liver.

Most forms of pathologic hyperplasia are caused by excessive hormonal or growth factor stimulation. E.g. after a normal menstrual period there is a burst of uterine epithelial proliferation that is normally tightly regulated by stimulation through pituitary hormones and ovarian estrogen and by inhibition through progesterone. However, if the balance between estrogen and progesterone is disturbed, endometrial hyperplasia will result.

Hyperplasia is also an important response of connective tissue cells in wound healing, in which proliferating fibroblasts and blood vessels aid in repair. Growth factors are produced by white blood cells (leukocytes) responding to the injury and by cells in the extracellular matrix. Hyperplasia can be associated with certain viral infections; papillomaviruses cause skin warts and mucosal lesions composed of masses of hyperplastic epithelium.

Hyperplastic process is a well-controlled process. If hormonal or growth factor stimulation abates (becomes less), the hyperplasia disappears. This sensitivity to normal regulatory control mechanisms distinguishes benign (Not life threatening) hyperplasia from pathologic proliferations like cancer. Pathologic hyperplasia constitutes a fertile soil in which cancerous proliferation may eventually arise. Patients with hyperplasia of the endometrium are at increased risk of developing endometrial cancer, certain papillomavirus infections also predispose to cervical cancers.

Atrophy: is shrinkage in the size of the cell by the loss of cell substance. When sufficient number of cells are involved, the entire tissue or organ diminishes in size, becoming atrophic. Although atrophic cells may have diminished function, they are not dead, it just represent a retreat by the cell to a smaller size at which survival is still possible. A new equilibrium is achieved between cell size and diminished blood supply, nutrition, or trophic stimulation.

Causes of atrophy: decreased workload (like immobilization of a limb to permit healing of a fracture), loss of innervations, diminished blood supply, inadequate nutrition, loss of endocrine stimulation and aging (senile/late life atrophy). Some of these stimuli are physiologic (like loss of hormone stimulation in menopause) and others are pathologic (like denervation).

Atrophy results from decreased protein synthesis and increased protein degradation in cells. Protein synthesis decreases because of reduced metabolic activity. The degradation of cellular proteins occurs mainly by the ubiquitin- proteasome pathway.

Nutrient deficiency and disuse may activate ubiquitin ligases, which attach multiple copies of the small peptide ubiquitin to cellular proteins and target these proteins for degradation in proteasomes. In many situations, atrophy is also accompanied by increased autophagy. Autophagy ("self-eating") is the process in which the starved cell eats its own components in an attempt to find nutrients and survive.

Metaplasia: is a reversible change in which one adult cell type (epithelial or mesenchymal) is replaced by another adult cell type. Cells sensitive to a particular stress are replaced by other cell types better able to withstand the adverse environment. Metaplasia is thought to arise by genetic "reprogramming" of stem cells rather than trans-differentiation of already differentiated cells. It may result in reduced functions or increased propensity for malignant transformation.

For example, the squamous change that occurs in the respiratory epithelium in habitual cigarette smokers. The normal ciliated columnar epithelial cells of the trachea and bronchi are focally or widely replaced by stratified squamous epithelial cells. Vitamin A deficiency may also induce squamous metaplasia in the respiratory epithelium. The "rugged" stratified squamous epithelium may be able to survive under circumstances that the more fragile specialized epithelium would not tolerate.

Barrett esophagus -In chronic gastric reflux, the normal stratified squamous epithelium of the lower esophagus may undergo metaplastic transformation to gastric or intestinal-type columnar epithelium. Metaplasia may also occur in mesenchymal cells but less clearly as an adaptive response. For example, bone is occasionally formed in soft tissues, particularly in foci of injury.

Inflammation

- Is Vascular response to injury
- Inflammatory cells, plasma proteins and fluid exit blood vessels and enter interstitial space
- Two types

1. Acute inflammation

- Edema and neutrophils in tissues is the common feature
- Common cause-
 - Infection- so that the acute inflammation can eliminate the pathogens
 - Tissue debris- helps clear the necrotic debris

Mediators of inflammation

A. Toll like receptors(TLR)

- ✓ Found on innate immune cells like macrophages and dendrites
- ✓ Pathogen associated molecular patterns(PAMP) which are located on microbes activate TLR

PAMP →TLR(e.g. CD14) up regulates→**NF-κB** (nuclear transcription factor)→activate immune response genes that lead to more immune mediators production

B. Arachnoid acid metabolites

- ✓ Cyclooxygenase produce prostaglandins

PGI₂, PGE₂, PGD₂ cause vasodilation and increase vascular permeability

- ✓ 5-lipoxygenase produce leukotriene

LTB₄ activate and attract neutrophils

LTC, LTD, LTE (collectively called slow-acting substance of anaphylaxis)-

They cause vasoconstriction, bronchoconstriction and increase vascular permeability

C. Mast cells

- ✓ Well distributed in connective tissues

- ✓ Activated by tissue trauma, complement proteins and IgE-antigen complex
- ✓ Cause immediate and delayed response

Immediate response release histamine (vasodilation of arterioles and increased vascular permeability) and the delayed response cause the production of arachnoid acid metabolites particularly the leukotriene

D. Complement system

- ✓ Complement proteins are proinflammatory proteins that circulate in blood and support the inflammation process
 - ✓ Circulate in blood in inactive form
 - ✓ Activated in three pathways
1. Classical pathway- this is due to antibody-antigen complex
C1 bound to IgG and IgM
 2. Alternative pathway- the microbial product itself cause the activation
 3. Mannose-binding lectin (MBL) pathway—MBL binds to mannose on microorganisms and activates complement.
- ✓ All pathways result in production of C3 convertase (mediates C3 → C3a and C3b), which, in turn, produces C5 convertase (mediates C5 → C5a and C5b)
 - ✓ C5b complexes with C6-C9 to form the membrane attack complex (**MAC**)
 - ✓ C3a and C5a (anaphylatoxins)—trigger mast cell degranulation, resulting in histamine-mediated vasodilation and increased vascular permeability
 - ✓ C5a—chemotactic for neutrophils
 - ✓ C3b is an opsonin (for phagocytosis)
 - ✓ MAC cause lysis of microbes by creating hole on them

Cardinal signs

1. Rubor (redness caused by dilation of vessels)
2. Dolor (pain due to increased pressure exerted by the accumulation of interstitial fluid
And mediators such as bradykinin and PgE2 cause pain by sensitizing nerve endings)
3. Calor (heat caused by increased blood flow)
4. Tumor (swelling due to an extravascular accumulation of fluid)
5. Fever

Pyogens (e.g., LPS from bacteria) cause macrophages to release IL-1 and

TNF, which increase cyclooxygenase activity in perivascular cells of the Hypothalamus and the increased prostaglandins raises **temperature set point**.

Neutrophils pathway to the injured tissue

A. Margination

Vasodilation slows blood flow in postcapillary venules and the leukocytes marginate from center of flow to the periphery.

B. Rolling

This is mediated by selectins that decrease the speed of the neutrophils

P- selectin: release from Weibel Palade bodies(found in endothelial cells) which is mediated by histamine.

E- selectin: induced by TNF and IL-1

Interaction between the neutrophils and selectin cause the rolling process

C. Adhesion

- cellular adhesion molecules (ICAM and VCAM) are upregulated on endothelium by TNF and IL-1
- Integrins are upregulated on leukocytes by C5a and LkB
- The interaction between the integrins and adhesion molecules result in firm adhesion of the neutrophils to the vessel wall

D. Transmigration and Chemotaxis

- Leukocytes transmigrate across the endothelium of 'post capillary venules and move toward chemical attractants (chemotaxis)

Chemotactic factors for neutrophils, produced at the site of injury, include:

1. Products from bacteria
2. Complement components, especially C5a and IL8
3. Arachidonic acid metabolites, especially leukotriene B4 (LTB4), kallikrein

E. Phagocytosis

Pathogens and necrotic tissues are engulfed and this is facilitated by opsonins such as **IgG and C3a**

F. Intracellular microbial killing

Oxygen-dependent microbial killing- effective than oxygen independent

- ✓ Phagocytosis initiates activity of the hexose monophosphate shunt, causing an oxidative burst and supplying electrons to an NADPH oxidase in the phagosomal membrane.
- ✓ One of the products of the NADPH oxidase reaction is superoxide anion which is further converted to hydrogen peroxide (H_2O_2) by dismutation.
- ✓ H_2O_2 oxidizes microbial proteins and disrupts cell walls.

Oxygen- independent microbial killing

occurs via enzymes present in leukocyte secondary granules (e.g., lysozyme in macrophages and major basic protein in eosinophils).

G. Resolution

Neutrophils undergo apoptosis and disappear within 24 hours after resolution of the inflammatory stimulus.

Macrophages

- ✓ Macrophages predominate after neutrophils and peak 2-3 days after inflammation begins.
 - ✓ Arrive to tissues with same steps as explained above
 - ✓ Ingest pathogens with the help of opsonins and destroy them via lysosome
 - ✓ **They manage the fate of acute inflammation**
1. Resolution and healing—Anti-inflammatory cytokines (e.g., IL-10 and TGF) are produced by macrophages.
 2. Continued acute inflammation—marked by persistent pus formation; IL-8 from macrophages recruits additional neutrophils.

3. Abscess—acute inflammation surrounded by fibrosis; macrophages mediate fibrosis via fibrogenic growth factors and cytokines.
4. Chronic inflammation—Macrophages present antigen to activate CD4 T-helper T Cells, which secrete cytokines that promote chronic inflammation.

Chronic inflammation

- It mostly occur de-novo without acute inflammation
- Causes are:
 1. Persistent infection (most common cause); 2. Infections with Viruses, mycobacteria, parasites, and fungi; 3. autoimmune disease;
 4. foreign bodies and some cancers
 - It occurs in two major patterns: chronic nonspecific inflammation and granulomatous inflammation.

Chronic nonspecific inflammation

- ✓ it is mediated by macrophage-lymphocyte interaction
- ✓ fibroblast and new blood vessel proliferation, scarring and distortion of tissue structures also occur
- ✓ cytokine production from macrophages activate lymphocytes and intern cytokine from lymphocytes activate macrophages
- ✓ B-lymphocytes get activated by macrophage antigen presentation and start antibody production

Granulomatous inflammation

Characterized by granuloma, which is a collection of epithelioid histiocytes
(Macrophages with abundant pink cytoplasm), usually surrounded by giant cells
And a rim of lymphocytes

Caseating and non-caseating granulomatous inflammation

Non-caseating: don't have central necrosis and Common etiologies
Include reaction to foreign material, sarcoidosis, beryllium exposure, Crohn Disease and cat scratch disease.

Caseating: central necrosis and are characteristic of tuberculosis and fungal infections

Wound healing

Healing begins when inflammation start, it involve regeneration and repair

Regeneration is replacement of damaged tissue with native tissue; dependent on regenerative capacity of tissue and it need an intact connective tissue for the parenchyma to regenerate

- Based on regeneration capacity there are three types of tissues

Labile tissue – have stem cells that continuously divide

- ✓ Small and large bowel (stem cells in mucosal crypts)
- ✓ Skin (stem cells in basal layer)
- ✓ Bone marrow (hematopoietic stem cells)

Stable tissue -replicate rapidly only in response to stimuli, $G_0 \Rightarrow G_1$

E.g. regeneration of liver by compensatory hyperplasia after partial resection. Each hepatocyte produces additional cells and then reenters quiescence.

- ✓ kidney, pancreas, endothelium, or mesenchymal cells (fibroblasts) also are stable tissues

Permanent tissue – no regenerative potential

- ✓ neurons, skeletal & cardiac muscles

Cellular proliferation – mediated by various growth factors and it is **the mechanism**

For repair and regeneration process

Platelet-derived growth factor (PDGF) - promotes the proliferative response of fibroblasts and smooth muscle cells on concurrent stimulation by other growth factors (so it indirectly results in collagen synthesis)

Epidermal growth factor (EGF) - promotes the growth of endothelial cells and fibroblasts, as well as epithelial cells.

Fibroblast growth factors (FGFs) - promote the synthesis of extracellular matrix protein (including fibronectin) by fibroblasts, endothelial cells, monocytes. It also promotes angiogenesis

Transforming growth factors (TGFs) – TGF functions similar to EGF.

Macrophage-derived growth factors (IL-1 and TNF) promote the proliferation of fibroblasts, smooth muscle cells and endothelial cells.

Repair

Repair is replacement of damaged tissue with fibrous scar

Occurs when regenerative stem cells are lost (e.g., deep skin cut) or when a tissue lacks regenerative capacity (e.g., healing after a myocardial infarction)

Granulation tissue

It is the first phase in the repair process. Consists of fibroblasts (deposit type III collagen), capillaries (provide nutrients), and myofibroblasts (contract wound)

The repairing process

1. Removal of debris begins in the early stages of inflammation and is initiated by liquefaction and removal of dead cellular material and other debris.
2. Formation of granulation tissue

Granulation tissue is highly vascular, newly formed connective tissue consisting of capillaries and fibroblasts; it fills defects created by liquefaction of cellular debris.

3. Scarring
 - a. Collagen is produced by fibroblasts. As the amount of collagen in granulation tissue progressively increases, the tissue becomes gradually less vascular and less cellular.
 - b. type III collagen will be replaced by collagen type I

Cutaneous healing – occurs by either primary or secondary intention

Primary intention - Wound edges are brought together (e.g., suturing of surgical incision); leads to minimal scar formation

Secondary intention - Edges are not approximated. Granulation tissue fills the defect; then contract the wound and scar will be formed

Delayed wound healing

1. Infection- example s.aureus
2. Vitamin C, copper, or zinc deficiency
- ✓ Vitamin C is an important cofactor in the hydroxylation of proline and lysine procollagen residues; hydroxylation is necessary for eventual collagen cross-linking.
- ✓ Copper is a cofactor for lysyl oxidase, which cross-links lysine and hydroxy lysine to form stable collagen.
- ✓ Zinc is a cofactor for collagenase, which replaces the type III collagen of granulation tissue with stronger type I collagen.

Dehiscence is rupture of a wound; most commonly seen after abdominal surgery.

Hypertrophic scar is excess production of scar tissue that is localized to the wound.

Keloid is excess production of scar tissue that is out of proportion to the wound.

1. Characterized by excess type III collagen
2. Genetic predisposition (more common in African Americans)
3. Classically affects earlobes, face, and upper extremities

Genetic disorders

1. mutation in single genes with large effects (mendelian disorders)

It is 80-85% familial. It could be either autosomal dominant or recessive

Autosomal dominant cases

If affected person marry unaffected person, the child will have 50% chance of being affected.

Examples

- A. Neurofibromatosis- it is mutation of gene on chromosome 17 which code for neurofibromin(NF1)

Features- Multiple neurofibromas on skin

Café au lait spots (flat pigmented spot on skin)

Pigmented iris hamartomas(Lisch nodules)

- B. Adult polycystic kidney- numerous bilateral cysts that destroy kidney
- C. Familial polyposis coli- numerous adenomatous polyps
- D. Achondroplasia- short limb with normal sized head and trunk
mutation on FGF receptor gene (on chromosome 4)
- E. Hereditary spherocytosis
- F. Familial hypercholesterolemia

Autosomal recessive cases

Unaffected carriers have 25% chance of having affected child (e.g. most inborn Error metabolism)

- A. Hereditary hemochromatosis-hemosiderin(Fe granules) accumulation in tissues along with organ damage and scarring
Note. Hemosiderosis is just hemosiderin accumulation in tissues especially macrophages without organ damage
 - Hemochromatosis have triad result: micro nodular cirrhosis , diabetes mellitus and skin pigmentation
- B. Sickle cell anemia- fibrotic spleen

X-linked recessive disorders

In males these disorder is expressed (males are hemizygous)

- A. Duchenne muscular dystrophy-its mutation in dystrophin gene which is located on the short arm of x chromosome
 - It result in variation and necrosis of muscle fibers
 - Progressive muscular weakness is the common feature
- B. Hemophilia
 - Hemophilia A – factor VIII deficiency
 - Hemophilia B – factor IX deficiency
 - Easy bruising and hemorrhage
 - Hemarthrosis (bleeding into a joint)

Single gene disorder without non classical inheritance

1. Tri nucleotide repeat mutation

Repetition is common in C or G nucleotide

- A. Fragile-x syndrome- is X-linked disorder in familial mental retardation gene(FMR-1) that is required for normal neural development
 - Is due to long repeating sequence of CGG(non- coding sequence)
 - It develop during oogenesis

Features: Mental retardation, a long face with a large mandible, large everted ears, & large testicles (macro-orchidism), flat feet, & low muscle tone.

- B. Huntington disease- CAG repetition

2. Mitochondrial genetic disorders

- Only the egg contribute mitochondria to zygote so only the mother can inherit this disorder to the child
- Organs with high energy requirement are mostly affected
- Symptoms progress with age
- E.g. Leber hereditary optic neuropathy => blindness

3. Genomic imprinting

Epigenetic changes occurring during gametogenesis mark some genes as either maternal or paternal origin. Different phenotypes occur depending on whether abnormal gene is of maternal or paternal origin.

- A. **Prader-Willi syndrome:** deletion of genes on the paternally inherited Chrom 15, while the maternal copy is silenced
 - short stature, hypotonia, profound hyperphagia, obesity, small hands & feet, & hypogonadism
- B. **Angelman syndrome:** deletion of genes on the maternally inherited Chrom 15, while the paternal copy is silenced
 - ataxic gait, seizures, and inappropriate laughter

Mental retardation is seen in both types

Chromosomal disorders

Aneuploidy: not exact multiple of 23

- non disjunction – trisomy

- lag of anaphase –monosomy
- polyploidy: due to diploid gamete or fertilization by 2 sperms
- triploidy

Down syndrome (trisomy 21)

Caused by maternal meiotic non disjunction and this is related to maternal age

Features: Mental retardation, typical flat facial profile, oblique palpebral fissures, and

Epicanthic folds

- Brushfield spot(small white spots on periphery of iris)
- Simian crease(single palmar crease)

Complication: Congenital heart disease, Leukemia, Increased susceptibility to infection, Alzheimer disease

Abnormalities of sex chromosomes

Klinefelter syndrome - karyotype 47, XXY is common variant and mostly maternal meiotic Non disjunction is the cause

Features: atrophic testes, tall stature, gynecomastia and decreased testosterone production

Mostly no association with mental retardation

Turner syndrome – complete or partial monosomy of X chr (45, X) with female hypogonadism

Features: streak ovaries (atrophic, fibrous strands, devoid of ova and follicles ovary)

- short stature, cystic hygroma of neck (leading to webbing), failure of 2^o sex characteristics, primary amenorrhea, infertility, congenital heart disease

Multifactorial disorders – genetics+environment

- Cleft Lip/palate, type II Diabetes Mellitus, autism, gout, hypertension, congenital heart disease, schizophrenia

Somatic cell genetic disorders

- Mostly cancer-causing mutations affect somatic cells(95%) ,not passed in germ line
- 5% of all cancers, mutations transmitted through the germ line, most are inherited in an autosomal dominant fashion
- E.g. retinoblastoma

Chronic myelogenous leukemia:translocation of a portion of the q arm of chromosome 22 to the q arm of chromosome 9, designated t(9:22)

Sex determination

- Genetic sex: presence/absence of Y chromosome
- Gonadal sex: histology of gonads
- Ductal sex: presence of derivative mullerian or wolffian ducts
- Phenotypic (genital) sex: appearance of the external genitalia

Examples of sexual ambiguity

True hermaphrodite: rare, presence of both ovarian & testicular tissues; 46XX (50%)

Pseudohermaphrodite: disagreement between phenotypic& gonadal sex

- **Female pseudohermaphrodite** has ovaries but ambiguous or virilized external genitalia: 46XX, 2^o to androgen exposure in congenital adrenal hyperplasia, autosomal recessive

- **Male pseudohermaphrodite** has testes but ambiguous/female-type external genitalia, 2° to defects in androgen; complete androgen insensitivity syndrome (testicular feminization) mutation in androgen receptor, X-linked recessive.

Cell Injury

Cell injury occurs when there is an injurious stimulus applied on the cell or if the cell became incapable to adapt to a stress or increased demand on it. If the cell adapts to a stress the cell will be capable to avoid injury. Listing all causes of cell injury could be tiresome, but they can be grouped as: oxygen deprivation, nutritional imbalances, physical agents, chemical agents, infectious agents, immunologic reactions, genetic defects and aging. Oxygen deprivation affects cellular aerobic respiration. Cell could get deprived of oxygen with a certain possible mechanisms: ischemia (loss of blood supply to tissues) and hypoxia (deficiency of oxygen in tissues). Hypoxia can be due to Cardio-respiratory failure, anemia or carbon monoxide poisoning.

Cellular response to injurious stimuli depends on the type, duration, and severity of the injury/stimulus. The structures and functions that are primarily affected by cellular injury are mitochondria (ATP production), cell membranes, protein synthesis, cytoskeletal integrity and genome.

Increased intracellular (cytosolic) calcium, oxidative stress and lipid peroxidation (membrane phospholipid damage by free radicals) can lead to mitochondrial damage. Mitochondrial damages in turn decrease generation of ATP molecules. Depending on the severity damage to the mitochondria may promote release of cytochrome c and other pro-apoptotic proteins from the mitochondria in to the cytosol leading to apoptosis. If the damage is severe the mitochondrial membrane permeability will be elevated greatly → loss of membrane potential → inability to produce ATP → necrosis. So both necrosis and apoptosis are possible outcomes.

The effects of ATP depletion is seen even when it decreases only by 5-10% from the normal amount. Classically we can see three consequences of ATP depletion, each having chains of consequences:

1. Failure of energy dependent ion pump (like that of sodium pump) → increased influx of calcium, sodium and as a consequence to the two solutes water in to the cells (calcium is the most important in mechanism of cell damage). This is coupled with efflux of potassium in to the extracellular fluid → endoplasmic reticulum swell, cell swell (as a result microvilli will be lost) and blebs (balloon like extensions from the cells)
2. Anaerobic glycolysis will be elevated to fill the gap in ATP demand as the aerobic pathway can't proceed because of unavailability of oxygen → increase lactic acid → decrease PH → decreased enzymatic activity and clumping of nuclear chromatin. Anaerobic metabolism is not an efficient process as it uses many energy sources to produce fewer energy compared to the aerobic, thus there will also be reduced glycogen in the cell.
3. Ribosomes will be detached because of the deficit in energy → decreased protein synthesis and those that are synthesized are likely to be misfolded.

Oxidative stress is another mechanism for cell injury and is mediated by free radicals. This will prevail if the free radical formation exceeds free radical neutralization by the antioxidant mechanism. Free

radicals are known to damage the various components of a cell, but the major consequences are lipid peroxidation primarily in membrane lipids, protein modifications and DNA damage.

Influx of calcium in to the cell is also a major damaging factor. If the intracellular calcium is very high it can lead to hyper-permeability of the mitochondrial membrane, which could lead to cell death. Calcium also activates a range of enzymes: ATPase (which decrease the amount of ATP), endonuclease (which can damage the chromatin), phospholipase and protease. Phospholipase and protease can damage membrane components, protease can also damage the proteins of the cytoskeleton. You should also know that a damage in the cytoskeleton will also mean a damage to the cell membrane much like a damage to the pillar of a house will mean a damage to the roof and possibly the walls.

Another mechanism of cell injury is through defects in membrane permeability. In this scenario, both the plasma membrane (cell membrane) and the membranes of the organelles can be affected. The mechanisms involved in the damage of membranes are

- Mitochondrial dysfunction → decreased ATP → decreased phospholipid reacylation/synthesis → phospholipid loss → membrane damage
- Increased intracellular calcium → phospholipase and protease activation → loss of phospholipid as well as membrane and cytoskeletal proteins → membrane and cytoskeletal damages
- Oxidative stress → lipid peroxidation → phospholipid loss → membrane damage
- Products of lipid breakdown by phospholipase (free fatty acids, lysophospholipids...)

An important point: a DNA damage and accumulation of misfolded proteins are usually followed by apoptosis while membrane damage is usually followed by necrosis (decreased ATP as a result of damage to mitochondrial membrane, by the loss of plasma membrane integrity and by the release of digesting enzymes because of damage to membrane of the lysosomes).

Cellular injury can have certain features: for example a cell injury may directly lead to apoptosis or it may wait until the reversible stage of the injury to cross the point of irreversibility to end up with necrosis. Other than this cellular injury can also lead to subcellular changes at the level of the organelles. When a cell injury is irreversible we can see a characteristic feature: i.e. extensive damage to all cellular membranes and inability to reverse mitochondrial dysfunction after resolution of original injury.

Consequences of cell injury depend on the type, state, and adaptability of the injured cell. For example striated skeletal muscle in the leg can withstand ischemia for 2 to 3 hours without irreversible injury, whereas cardiac muscle dies after only 20 to 30 minutes. Consequences of cell injury depend on the type, state, and adaptability of the injured cell. Genetically determined diversity in metabolic pathways are also important in determining the consequences.

In cell injury the morphologic changes are delayed and are usually preceded by the loss of functional loss. This is because the cells could stop undergoing their normal activities when the conditions around them are altered even if the alteration is minimal, but to cause morphologic change a relatively larger alteration (for example larger intensity or frequency of stimuli) is required. So there will be early loss of cellular function but delayed changes in morphology are seen in cellular injuries. Cell injury, particularly the reversible one give the following common changes in morphology:

1. **Cellular swelling** (aka. hydropic change or vacuolar degeneration) – this is the first manifestation for almost all forms of injury to cells. Grossly we see pallor, increased turgor, and increase in weight of the organ. Microscopically we see small and clear vacuoles within the cytoplasm.
2. **Fatty change** – this one is principally encountered in cells participating in fat metabolism and is characterized by appearance of lipid vacuoles in the cytoplasm.
3. **Ultra-structural changes** – this one shows many changes that seems to lead to necrosis like:
 - Plasma membrane blebbing - as a result we may see blunting or distortion of microvilli and loosening of intercellular attachments.
 - Mitochondrial changes - particularly swelling
 - Dilation of the endoplasmic reticulum - as a result we may see detachment of ribosomes and dissociation of polysomes.
 - Nuclear alterations

Even after all these changes the cell may recover and become normal since the injury is still reversible. But if the cell crosses the point of irreversibility it will surely undergo necrosis and not even apoptosis.

When cells are exposed to sub-lethal or chronic stimuli they will show **subcellular alterations**. This alterations can be summarized by the four major changes that are seen in the organelle level:

- **Lysosomal catabolism** – can be due to hereditary lysosomal storage disorders (deficiencies of enzymes) or else it could be due to drug induced (acquired) lysosomal diseases. A good example for drugs that can induce lysosomal diseases is **chloroquine**.
- **Induction (hypertrophy) of Smooth endoplasmic reticulum** – for instance, barbiturates are metabolized in the liver by the cytochrome P-450 mixed-function oxidase system found in the SER. Protracted use of barbiturates leads to a state of tolerance, with a decrease in the effects of the drug and the need to use increasing doses. This adaptation is due to increased volume (hypertrophy) of the SER of hepatocytes and increased P-450 enzymatic activity.
- **Mitochondrial Alterations** - alterations can be in the number, size, shape or function of mitochondria. It may assume extremely large and abnormal shapes (**megamitochondria**), as seen in hepatocytes in various nutritional deficiencies and alcoholic liver disease.
- **Cytoskeletal Abnormalities** – can occur in a variety of pathologic states. These abnormalities may be manifested as an abnormal appearance and function of cells (like **hypertrophic cardiomyopathy** and **immotile cilia syndrome**), aberrant movements of intracellular organelles, defective cell locomotion, or intracellular accumulations of fibrillar material as in alcoholic liver disease.

Cell Death

Now we know that alteration in cellular morphology always accompany cell death. But this changes are not the same for the two major types of cell death: necrosis (Greek meaning the process of death) and apoptosis (Greek meaning falling off or away).

In necrosis the following cellular changes are implicated:

- Dilatation of endoplasmic reticulum and loss of ribosomes from rough endoplasmic reticulum and marked dilation of mitochondria with the appearance of large amorphous densities.
- 'Blebbing' from the cell membrane and discontinuities in plasma membrane and organelle membranes.
- Myelin figures – these are whorled phospholipid masses that are derived from damaged cellular membranes.
- Increased eosinophilia – due to two mechanisms (the denatured proteins formed due to the damage have increased eosin binding and there is a loss of RNA which was basophilic).
- Glassy homogeneous appearance
- Nuclear change of three steps: Pyknosis (nuclear shrinkage and increased basophilia) → Karyorrhexis (fragmentation of pyknotic nucleus) → Karyolysis (fading of basophilia)

Tissue necrosis is always pathological unlike apoptosis which can be either pathologic or physiologic. The causes of necrosis are different and accordingly the types of necrosis are also different. The patterns (types) of necrosis are shown below:

1. Coagulative necrosis – in this type of necrosis the basic tissue architecture is preserved for at least several days. This form is seen in infarcts (areas of ischemic necrosis) of solid organs like the liver.
2. Liquefactive necrosis – this type involves a transformation of the tissue into a liquid viscous mass. This type is usually seen in focal bacterial infections and hypoxic death of cells within the CNS.
3. Caseous necrosis – the term "caseous" (cheese-like) is derived from the friable yellow-white appearance of the area of necrosis. It is most often encountered in foci of tuberculous infection. Microscopically you can see granular eosinophilic mass of fragmented or lysed cells with the total loss of cell detail.
4. Fat necrosis – this involves focal areas of fat destruction. It could be enzymatic fat necrosis (pancreatic enzymes liquefy the membranes of fat cells in the peritoneum, split the triglyceride esters contained within fat cells → the released fatty acids combine with calcium to produce grossly visible, soft, chalky white areas (fat saponification). It is not always enzymatic and it can also be traumatic. Microscopically we see shadowy outlines of necrotic fat cells with basophilic calcium deposits; with inflammatory reaction.
5. Fibrinoid necrosis – usually seen in immune reactions involving blood vessels leading to immune complex deposition in the walls of arteries. The inflammatory deposits join hands with fibrin that has leaked out of vessels to result in a bright pink and amorphous appearance.
6. Gangrenous necrosis – this is a descriptive term for tissue that has lost its blood supply and has undergone coagulative necrosis involving multiple tissue layers. There is also a special term 'wet gangrene', used when there is a superimposed bacterial infection to the gangrene where liquefactive action of the bacteria and leukocytes have liquefied it.

When cells die with necrosis there will be leakage of intracellular proteins into the circulation and this provides a means of detecting tissue damage using blood or serum samples. For example: elevated creatine kinase and troponin is indicative of myocardial infarction.

Apoptosis is the other mechanism of cell death. Typically, apoptosis is a physiologic mechanism but at times it is associated with pathologic scenarios.

The physiologic mechanism of apoptosis is also known as programmed cell death. This name indicates that the cells undergoing apoptosis were meant to die so that the physiology of the organ would be maintained. Apoptosis when occurring on a physiologic basis have roles in: **embryogenesis, maintenance of a steady number of cells in tissues, involution of hormone-dependent tissues** upon hormone deprivation (a typical example is the case with **menstruation**), **elimination of potentially harmful lymphocytes** (to avoid auto immune conditions), **removal of cells that have served their useful purpose** and cytotoxic T cell mediated immunity (where the **infected cells die with an apoptosis mediated by cytotoxic T cells**). Apoptosis is known to have a preventive effect against neoplastic conditions and autoimmune disorders. But it **increases the risk for neurodegenerative diseases** as nerve cells do not restore their number after a portion of their population is cleared by apoptosis.

At times apoptosis can have pathologic implications: **DNA damage, accumulation of misfolded proteins, viral infections, atrophy, tumors and ischemia.**

Apoptosis have **two phases: initiation and execution**. The **initiation phase** involves **activation of procaspases in to the active caspases by either intrinsic or extrinsic mechanisms**. The **execution phase** corresponds to the **action of the caspases to bring about apoptotic cell death**.

Caspases are activated through either **extrinsic or intrinsic routes**:

- The extrinsic pathway is **initiated when a ligand is provided from the extracellular environment (like fas ligand of killer lymphocytes binding to fas receptor) → this facilitates the binding of adapter proteins to the fas proteins in the cell membrane → the adapter protein then binds procaspase 8 on its other side → this initiates aggregation of procaspases 8 proteins → procaspase 8 is then cleaved to caspase 8 which initiates caspases cascade leading to apoptosis.**
- The intrinsic pathway is **initiated when the cytochrome c originally contained in the inter-membrane space of the mitochondria is released in to the cytoplasm → the cytochrome c binds to Apaf-1 (an adaptor protein) → the Apaf-1 aggregates allowing the binding of procaspase 9 → the procaspases 9 is activated to caspases 9 → caspase cascade activated leading to apoptosis.**

One major difference between the intrinsic and extrinsic pathways is that the **intrinsic activates procaspase 9** while the **extrinsic activates procaspase 8**. Apoptosis **can also be initiated by the perforin/granzyme pathway of cytotoxic T cells**. The perforin from the T cells can perforate the cell membrane of the cell and enter the cytoplasm giving granzyme A and B. It is granzyme B that is involved in initiating apoptosis by activating procaspase 10 in to caspase 10. The stage that is common to all the pathways is the **activation of caspase 3** by caspase 9, caspase 8 and caspase 10 of the respective pathways.

A **procaspases is activated in to a caspase by cleavage**. Caspases will always have **two subunits (one large and one small subunits)**. In the caspases cascade one caspase can activate many caspases in the next level of the cascade.

In apoptosis there is clearance of apoptotic cells by phagocytes (phospholipids, i.e. phosphatidylserine are expressed on outer surface of cell membrane acting like an identification mechanism by the phagocytes). As a result there is no inflammation in apoptosis because the cellular contents are not released into the surrounding but into apoptotic bodies. But in necrosis there is enzymatic digestion and leakage of cellular contents so there is frequent adjacent inflammation. This is not the only difference between necrosis and apoptosis. We can also have other differences:

- In necrosis the cell size increases by swelling but in apoptosis cell shrinks (decrease in size)
- In necrosis the nucleus follows changes in steps (Pyknosis → karyorrhexis → karyolysis) but in apoptosis the nucleus is simply reduced to nucleosome sized fragments
- The plasma membrane in necrosis is disrupted but in apoptosis although it may show altered structures (especially in orientation of lipids) it will still be intact
- Necrosis is invariably pathologic indicating an irreversible cell injury, on the other hand apoptosis is often physiologic (means of eliminating unwanted cells) but at times pathologic (after some forms of cell injury, especially DNA damage)

Pathologic calcification

Pathologic calcification can be of two types, metastatic or dystrophic.

- Metastatic calcification – related to hypercalcemia and occurs in viable (normal) tissue. The causes of the hypercalcemia can be increased PTH (parathyroid hormone), destruction of bone tissue, vitamin D disorders or renal failure. Areas of the body that are prone to this kind of calcification are internal alkaline compartment and interstitial tissues (gastric mucosa, kidneys, lungs, systemic arteries and pulmonary veins). There may be no clinical manifestation (could be asymptomatic) and if symptoms are present the commonest are: respiratory insufficiency and renal damage.
- Dystrophic Calcification – occurs in non-viable tissue (necrotic areas). It occurs in two phases: initiation (nucleation) and propagation. The initiation phase depends on mitochondria of dead/dying cells, membrane bound vesicles derived from degenerating cells and concentration of calcium. The propagation phase depends on concentration of calcium & phosphate as well as presence of inhibitors and other proteins in the extracellular space. In this type of calcification we see formation of crystalline calcium phosphate mineral in a form that is similar to the hydroxyapatite of bone. The tissue may undergo ossification and can cause organ dysfunction. Grossly we see fine, white granules or clumps with gritty appearance. Microscopically, basophilic, non-crystalline amorphous deposits and hydroxyapatite crystals. A typical microscopic feature of the dystrophic form is the psammoma bodies which are concentric lamellated calcified structures.

Oxidative stress and antioxidant system

In chemistry, a free radical is any atom, molecule, or ion with one or more unpaired valence electrons. In order to be stable the free radicals must take electron from atoms/molecules. This means they are oxidants. But not all oxidants are free radicals. For example: H_2O_2 (hydrogen peroxide), HOCl

(hypochlorous acid), O_3 (ozone), 1O_2 (singlet oxygen), $ONOO^-$ (peroxynitrite) and Me^{n+} (transition metals) are strong oxidants but not free radicals. Once a free radical succeeds in gaining electron from a nearby molecule, it leaves its victim short an electron and has now made this new molecule a free radical, which will in turn, try and steal an electron as well. The result is what we call a free radical cascade, an enormous chain reaction of free radicals that quickly wreaks havoc on living tissue. It is estimated that the chain reaction can trigger 6.023×10^{21} billion molecule to react per second! It is important to study free radicals in medicine because they can be the soul contributors for more than 100 pathologic conditions (examples - accelerated aging, cancer, heart attacks, stroke and arthritis). But free radicals have certain advantages, but in controlled level. For example they can kill harmful microorganisms in immune cells.

Most free radicals in biological systems are derivatives of oxygen (Reactive Oxygen Species, ROS), but there are also derivatives of nitrogen (Reactive Nitrogen Species, RNS), Reactive Metabolites or Intermediates. Free radicals can be R_3C (carbon-centered), R_3N (nitrogen-centered), $R-O$ (oxygen-centered) or $R-S$ (sulfur-centered). Free radicals are highly reactive $\rightarrow$ Generate new radicals by chain reaction $\rightarrow$ Cause damage to biomolecules, cells and tissues. But they have very short half-life.

The half-life of free radicals differ from one type to another. For example the half-life of some free radicals like hydrogen peroxide is measured in minutes while that of hydroxyl radicals is in nanoseconds. Free radicals undergo the chemical reactions indicated below:

Addition: $R\cdot + H_2C=CH_2 \rightarrow R-CH_2-CH_2\cdot$

Hydrogen abstraction: $R\cdot + LH \rightarrow RH + L\cdot$

Electron abstraction: $R\cdot + ArNH_2 \rightarrow R^- + ArNH_2^+$

Termination: $R\cdot + Y\cdot \rightarrow R-Y$

Disproportionation: $CH_3CH_2\cdot + CH_3CH_2\cdot \rightarrow CH_3CH_3 + CH_2=CH_2$

The sources of free radicals can be endogenous or exogenous. The endogenous sources are from the different organelles of the cells through leaking (mitochondria, lysosome, peroxisome, endoplasmic reticulum and microsomes) and release (nitric oxide from certain immune cells). Different enzymes are directly involved in the production of reactive species (example – nitrogen oxide synthase, NADPH oxidase and xanthine oxidase). The exogenous sources are radiations and chemicals (example - chemicals that causes the release of iron).

Some of the factors that promote the formation of free radicals are smoking, drinking alcohol, industrial pollutions, herbicides/pesticides and high fat diet.

Our body constantly reacts with oxygen during breathing and our cells produce energy. As a consequence of this activity, highly reactive molecules are produced known as free radicals. Free radicals can also be produced by other mechanisms including exogenous sources (example – radiation and certain drugs). Free radicals interact with other molecules within cells. This can cause oxidative

damage to proteins, membranes and genes. Oxidative damage has been implicated in the cause of many diseases, such as cancer and Alzheimer's and has an impact on the body's aging process. Lipids, proteins and genetic materials are most susceptible to oxidative damage by ROS.

The lipid bilayer membrane is composed of a mixture of phospholipids and glycolipids that have fatty acid chains. Lipids are a major component of living organisms and probably the first easy target of free radicals once they are produced, lipid peroxidation might play an important role in initiating and/or mediating some aspects of the aging process. It has been widely demonstrated that there is an age-associated increase in the steady-state concentrations of lipid peroxidation products.

- Initiation - the production of $R'/PUFA$ radical/ ROO' by the interaction of PUFA with free radicals generated by other means. R' and ROO' are degraded to malon dialdehyde - an indicator of fatty acid break down by free radicals.
- Propagation - one free radical generates another free radical in the neighboring molecules → chain reaction → destruction of fine architecture & integrity of the membranes.
- Termination - reactions in membrane lipids are terminated when the carbon or peroxy radicals cross-link to form conjugated products that are not radicals.

Lipid peroxy radicals react with other lipids, proteins, and nucleic acids; propagating thereby the transfer of electrons and bringing about the oxidation of substrates. Cell membranes, which are structurally made up of large amounts of PUFA (polyunsaturated fatty acids), are highly susceptible to oxidative attack and after an attack there will be breakage of chains in the target lipid. Consequently, changes in membrane fluidity, permeability, and cellular metabolic functions result.

Lipid peroxidation at different levels have different outcome. If the level is low stress signaling will be induced giving rise to antioxidant induction thus cell survival. If the level is moderate the stress signaling will induce apoptosis. If the level of lipid peroxidation is high, there will be gross structural and metabolic damage with membrane lysis leading to necrosis.

Oxidative attack on proteins results in site-specific amino acid modifications, fragmentation of the peptide chain, aggregation of cross-linked reaction products, altered electrical charge and increased susceptibility to proteolysis. Sulphur containing amino acids, and thiol groups specifically, are very susceptible sites. The oxidation of iron-sulphur centers by superoxide destroys enzymatic function. Thus it destroys the structure, functions of essential proteins and enzymes and whole cell metabolism is blocked. One of the factors that increase the susceptibility of proteins for ROS is glycation by carbohydrates. With an oxidative attack a protein will be more susceptible for aggregation (cross linking with other oxidized proteins) and proteolysis. The electrical charge will also deviate from the normal and there will be enzyme inactivation.

In the process of cataractogenesis, oxidative modification plays a significant role in cross-linking of crystalline lens protein, leading to high-molecular-weight aggregates, loss of solubility, and lens opacity. Lipofuscin, an aggregate of peroxidized lipid and proteins accumulates in lysosomes of aged cells, brain cells in Alzheimer's disease, and iron overloaded hepatocytes.

Activated oxygen and agents that generate oxygen free radicals, such as ionizing radiation, induce numerous lesions in DNA that cause deletions, mutations and other lethal genetic effects. Degradation of the base will produce numerous products, including 8-hydroxyguanine, hydroxymethyl urea, urea, thymine glycol, thymine and adenine ring-opened and -saturated products. Characterizations of this damage to DNA has indicated that both the sugar and the base moieties are susceptible to oxidation, causing base degradation, single strand breakage, and cross-linking to protein. Consequences of DNA oxidation include DNA adducts/AP sites/Strand breaks → mutation → cancer. The other notable consequences are breakage of strands, removal of nucleotides, modification of bases and cross linking with proteins. But it can also give rise to DNA repair mechanism which can deplete energy, activate other signaling pathway, express error prone polymerase or lead to imbalanced induction of DNA repair enzymes.

Mutation arising from selective modification of G:C sites specially indicates oxidative attack on DNA by ROS. The action of 8-oxodeoxy- guanosine as a promutagen, as well as in altering the binding of methylase to the oligomer so as to inhibit methylation of adjacent cytosine has been reported in cases of cancer development. ROS have also been shown to activate mutations inhuman C-Ha-ras-1 protooncogene, and to induce mutation in the p53 tumor-suppressor gene.

ROS may interfere with normal cell signaling, resulting thereby in alteration of the gene expression, and development of cancer by redox regulation of transcriptional factors/activator and/or by oxidatively modulating the protein kinase cascades. ROS also induce various early response or stress-response genes like c-fos, c-jun, jun-B, jun-D, c-myc, erg-1, and heme oxygenase-1. The oxidative damage of mitochondrial DNA also involves base modification and strand breaks, which leads to formation of abnormal components of the ETC (electron transport chain). This results in the generation of more ROS through increased leakage of electrons, and cell damage. Oxidative damage to mitochondrial DNA may promote cancer and aging.

Free radicals, at times like immune system functions can be considered to be useful but if their amount is higher than a threshold they can predispose to many life induced diseases. Especially the brain is uniquely sensitive for damage by free radicals (intolerant for blood flow interruption, limited regeneration, aging sensitive, high amount of PUFA, circuit based function in which the little damage expands fast and calcium dependent processes). Abnormal protein aggregates and inflammation can also prevail together with other outcomes of oxidative damages in the brain. For example ROS and RNS which are produced in inflammation or injury are involved in the pathogenesis of Alzheimer's disease. In Alzheimer's disease and stroke abnormal cholesterol metabolism, oxidative processes and vascular lesions are apparent. There are more and more mutations even in normal aging → Elevation of beta-amyloid → beta-amyloid polymerization → Production of ROS → Lipid peroxidation → Tau protein formation → Neuronal dysfunction (Alzheimer's). Anatomically Alzheimer's can show two proteins contributing to degeneration:

- Extra-neuronal plaques: by accumulation of beta-amyloid proteins. In normal brain this proteins would have been broken down and eliminated.

- Intra-neuronal neurofibrillary tangles: by accumulation of insoluble twisted fibres (tau proteins). In normal brain these proteins produce microtubules but in case of this pathology the microtubule structure collapse.

Parkinson's disease also results from oxidative damages: Increased hydrogen peroxide → increased dopamine oxidation → decreased glutathione in substantia nigra → increased iron in substantia nigra → formation of hydroxyl radical → lipid peroxidation increases → Parkinson's disease results

Atherosclerosis also results from oxidation of low density lipid in the vessels. The macrophages will take this oxidized lipids and convert to foam cells → coagulate to form fatty streak (yellow patches in arteries) but when they die they will release the fat contained in them. Oxidized LDL will also promote cell damage, adhesion of monocytes to endothelium, pro-coagulant properties of vascular cells, migration and proliferation of smooth muscle and blockage of NO (a vasodilator) in the vessels.

Oxidative stress can also promote aging of cells. Once the mitochondria ages it can't perform in an effective manner and more over it produces and leaks larger amount of ROS.

Free radicals can cause cancer by directly oxidizing thus damaging the genetic material and by inhibiting antioxidant mechanisms. So redox mechanism can be a primary cause of cancer.

There should always be balance between reactive species and antioxidants. If the antioxidant mechanism is in balance with the oxidative mechanism the damage will be minimal but if the antioxidant is deficient oxidative damage takes the upper hand. Antioxidant are compounds that protect cells from the damage caused by oxidation. This is achieved by preventing electron transfer from oxygen to organic molecules, stabilizing free radicals and terminating free radical reactions. There are two types of antioxidant classifications:

According to their location

- Plasma antioxidants: Ascorbic acid (Vitamin C), bilirubin, uric acid, transferrin, ceruloplasmin, β -carotene
- Cell membrane antioxidants: α -tocopherol (Vitamin E)
- Intracellular antioxidants: Superoxide dismutase (SOD), catalase, glutathione peroxidase (GPx)

According to their nature and action

- Enzymatic antioxidants: SOD, catalase, GPx, glutathione reductase
- Non-enzymatic antioxidants: nutrient or metabolic

Nutrient: β -carotene, α -tocopherol, ascorbic acid,

Metabolic: Bilirubin, uric acid, ceruloplasmin, ferritin, transferrin, albumin, glutathione

Superoxide dismutase (SOD): found almost in every cell in 3 metalloenzyme (with different prosthetic groups) forms: copper-zinc in cytoplasm with two subunits, manganese (Mn) in mitochondria and copper in extracellular space. It is the first line defense against super oxide: $2O_2^{\cdot -} + 2H^+ \rightarrow H_2O_2 + O_2$

Catalase: iron dependent enzyme found all over the body but especially concentrated in the liver and red blood cells: $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$

Glutathione peroxidase (GPx): selenium dependent enzyme. The entire process is driven by energy production at the cellular level, which involves proper thyroid hormone levels, healthy mitochondrial function, and an active pentose-phosphate metabolic pathway.

α -tocopherol (vitamin E): the most important lipid soluble antioxidant. Found in all cell membranes and also in adipose tissues preventing the peroxidation of membrane phospholipids and avoiding cell membrane damage.

Ascorbic acid (vitamin C): water soluble antioxidant present in citrus fruits, potato, tomato and green leafy vegies. It promotes the regeneration of vitamin E. It is also a reducing agent (electron donors) and scavenges free radicals decreasing lipid peroxidation. It is generally a chain breaking antioxidant.

Carotenoids: 40 carbon chains joined in double bonds. Show high light absorption thus appearing bright (red or orange). They occur as pigments in certain bacteria, algae and higher plants like carrot root. β -carotene is the most important carotenoid and is composed of 2 molecules of vitamin A (retinol). It is converted to retinol in the intestine and afterward it will be able to quench singlet oxygen. This ability to quench singlet oxygen is why it is used in the treatment of erythropoietic protoporphyria. Lycopene is another carotenoid and is responsible for pigments of certain fruits and vegetables like tomato but also have antioxidant property.

α -lipoic acids: vitamin like compound from animal and plant source and also produced in the human body. It plays a key role in recycling other important antioxidants such as ascorbic acid, α -tocopherol and glutathione.

Other important nutrient antioxidantsSource

1. Coenzyme Q₁₀organ meats (best heart), beef, chicken
2. Seleniumsea foods, meats, whole grains
3. Proanthocyanidinsgrape seeds
4. Catechinsgreen tea
5. Quercetinonions, red wine, green tea

Glutathione (GSH): it is a substrate for GSH redox cycle. It is also an antioxidant as it scavenges hydroxyl

Other metabolic antioxidants	Purpose
1. <u>Uric acid</u>	scavenge singlet oxygen and hydroxy radical
2. <u>Ceruloplasmin</u>	inhibit iron and copper dependent lipid peroxidation
3. <u>Transferrin</u>	prevents iron-catalyzed radical formation
4. <u>Albumin</u>	scavenge radicals on its surface
5. <u>Bilirubin</u>	protects albumin bound FFA from peroxidation

radicals and singlet oxygen. GSH is an important component in xenobiotic metabolism.

But you should always remember that supplying a required substance through food is better than using drugs. For example you could prefer to give the important fruit or vegetable instead of vitamin supplements.

One of the best indication as an antioxidant is green tea. Its antioxidant power is 200 times greater than that of vitamin E. green tea protects against GIT and respiratory infections but more importantly decreases the risk for certain cancers (colon, pancreas and stomach), reduce the risk for heart disease, LDL peroxidation and RBC lysis through its high free radical scavenging potential.

The best ways to promote the minimization of free radical formation is decreasing caloric intake and adapting a life with routine physical exercise.

What advice should a physician give?

Don't smoke, decrease exposure to the sun, don't over consume alcohol, avoid trans fats or hydrogenated oils, get your cardio exercise (sprinting or interval training), consume antioxidant rich foods and use a good antioxidant formula and keep your stress levels down.

Microbial structure and Function

Microbiology studies all organisms that can't be seen in the naked eye but with the microscope. Medical microbiology in contrast deal with the microorganisms that cause human disease or those that are found as normal micro-biota that lives with a commensal or mutual relationship.

The type of bacteria that is important in medicine is the eubacteria. It is surrounded by cell wall with complex carbohydrates. Have peptidoglycan (murine/muramic acid). It indifferently have a cell membrane but some have a double cell membrane. Some are motile with flagella. Since they are prokaryotic cell their nucleus or organelles are not enveloped. There are three morphological types of bacteria:

- Coccus (cocci): spherical
- Bacillus (bacilli): rod shaped or cylindrical
- Spiral (helical): have subtypes (spirillum, spirochaetes, comma shaped)
- Other than the 3, intermediate shapes like coccobacilli, fusiform...

The arrangement of the cells or the cell grouping is dependent on the plane of successive cell division. For example the cocci can be: micrococcus (single), diplococcus (duplicated), streptococcus (chain), staphylococcus (cluster), tetrad (packet of four) and sarcinae (cubical packet of eight). The bacilli can also be single bacilli, diplobacilli, streptobacilli, V shaped or palisade (looks like a wall of bacilli). A bacterial cell includes the following structures:

- External (appendages): flagellum (Rapid motility, chemotaxis, sensing), pili (Adhesion, sex), fimbriae, capsule (anti-phagocytosis)
- Envelope: cell wall (maintenance of shape/rigidity), cytoplasmic membrane (rapid permeability)
- Internal (cytoplasm): nucleoid (nuclear material), mesosome (cell division), ribosome (protein synthesis)
- Structural changes: endospore (resting but resistant state only in few types of bacteria)

Flagella is the organ of motility for motile bacteria. It is made of helical protein filaments of uniform length. It also expresses a flagellar antigen (H antigen). The flagella consists of three basic parts (filament, hook and basal body). Filament is the outermost region which contains the globular protein flagellin. But unlike the eukaryotic filaments this is not covered by a sheath. The hook is the wider segment that anchors filament to basal body. The basal body in turn is a complex structure with a central rod surrounded by a set of rings. Atrichous (aflagellate) bacteria are those without flagella. Flagellated bacteria can be Monotrichous (single polar flagellum), Lophotrichous (2 or more at 1 end/tuft at one end), Amphitrichous (tufts at both polar ends) and Peritrichous (distributed around the cell, E.g. Enterobacteria).

Endoflagella (Periplasmic flagella/axial filament) is a special type of flagella seen in spirochetes (like *T. pallidum*, *Borrelia burgdorferi*). This type of flagella is membrane covered and its rotation produces corkscrew type of motion.

Pilli are hair-like appendages that are shorter, straighter and thinner than flagella. These are particularly seen in many gram negative bacteria but also in some gram positive. Example can be in urinary and gastro intestinal bacteria like *N. gonorrhea*, *E. coli*, *S. flexneri*, *P. aeruginosa* and in the oropharynx like *S. pyogenes*. Pilli is not used for motility but primarily for adhesion (attachment) to host cells. F-pilus (sex pili) is used for genetic transfer (conjugation). Pilli also increase the buoyancy of the bacteria enabling it to float on the surface of water where there is more oxygen. According to the functional significance pilli is categorized into adhesins, lectins, invasins and sex pili.

Glycocalyx is the capsule and the slime layer together. The slime layer is the thin polysaccharide substance loosely attached to the cell wall. The capsule is still polysaccharide layer but its attachment to the cell wall is rather firm.

The capsule is gelatinous, polysaccharide (except in *B. anthracis*) in nature. Poorly stained & defined material which inhibits phagocytosis of the bacteria. Capsulated strains form mucoid/smooth (S) colonies. Non-capsulate strains form rough (R) colonies. Change from Smooth to Rough colonies (through mutation) correlates with loss of virulence. Other than inhibiting phagocytosis the polysaccharides will also stop the complements of immune system as they can't penetrate sugars. The capsule also prevents drying out and in some instances used for adherence (*Streptococcus mutans* to enamel of teeth). Encapsulated bacteria: *Streptococcus pneumoniae*, *Klebsiella*, *Streptococcus*, *Neisseria meningitidis*, *Haemophilus*, *Yersinia* ...

The cell envelope contains the cell wall and the cytoplasmic membrane.

The cell wall is a strong basic frame which gained its strength from the peptidoglycan (muropeptide/murine). The peptidoglycan is a complex polymer of amino-sugar. The Peptidoglycan which is the back-bone of the cell is made up of 2 sugars - N-acetyl muramic acid (NAM) & N-acetyl glucosamine (NAG) attached by β -1-4 glycosidic linkage. Lysozyme (Muramidase) breaks this linkage cleaving it to a disaccharide units. The cell wall is the point of distinction between gram positive and negative bacteria.

Components of Gram positive bacterial cell wall include peptidoglycan, polysaccharides (50 - 80%, neutral sugars) and teichoic/teichuronic acid (about 50% of the weight of the cell wall).

The difference teichoic acid and teichuronic acid resides in the abundant phosphate in teichoic acid. Teichoic acid contains ribitol or glycerol residues and is associated with lipids (Lipoteichoic acid which anchors some portion of the membrane). Teichoic acid has the following actions: sequester/bind cations like Mg^{2+} , serve as attachment site for bacteriophages or serves as source of major surface antigens.

M-proteins are found in the cell wall of Group A *Streptococcus* (*S. pyogenes*). It helps in adherence/colonization of the bacteria and prevents the cell from antibiotics or other harmful substances.

Components of Gram negative cell wall includes lipoprotein (mediates the attachment of Outer Membrane to Peptidoglycan), outer membrane (phospholipid-protein bilayer), peptidoglycan (2-10%) and lipopolysaccharide (LPS). The LPS contains 3 parts:

- a. O-specific chain (somatic Ag), a repeating unit: highly species specific, contain antigenic determinant
- b. Core polysaccharide: has KDO (2-ketone-3-deoxyoctanoic acid) & amino sugars
- c. Lipid A (Toxic, Endotoxin) made up of long chain fatty acids

Property	Gram Positive (+ve)	Gram negative (-ve)
Thickness of wall	Thick (20-80 nm)	Thin (10 nm)
Number of layers in wall	1	2
Peptidoglycan content	50% or even more	10-20%

Teichoic acid in wall	Yes	No
Lipid & lipoprotein content	0-3%	58%
Protein content	0%	9%
Lipopolysaccharide	0%	13%
Sensitive to penicillin	Yes	Less sensitive
Digested by lysozyme	Yes	Weakly

There are cell wall deficient bacteria. The first is the pleomorphic bacterial family of mycoplasma. Another is the L-form bacteria produced in the lab or the body by penicillin treatment and it replicates in ordinary media. Spheroplast is another cell wall deficient bacteria, although it retains some residue of the cell wall but a nonfunctional one. Spheroplasts are produced from gram negative bacteria artificially by lysozyme or growth with penicillin or other agents. Spheroplasts are osmotically fragile and thus less stable than the L-form. Protoplast is osmotically fragile and produced with lysosome or hypertonic medium but the precursor cell is gram positive not gram negative. Spheroplasts and protoplasts must be put in osmotically protected medium. If the antibiotic or the other stimulus is withdrawn cell wall deficient bacteria may grow it back.

The cytoplasm is surrounded with the semipermeable cytoplasmic membrane which is composed of a bilayered phospholipid and proteins. Its function to selectively allow solutes in to cell, excretion of hydrolytic exoenzymes, holding some enzymes and carrier proteins, synthesize cell wall (macromolecular synthesis) and electron transport and oxidative phosphorylation. Cytoplasm contains convoluted invaginations for cell division often at site of septum formation but also at lateral locations, we call them mesosomes and are common in gram positive. The mesosomes can be septal or lateral. They can be classified as vesicular, lamellar or tubular.

The cytoplasm of prokaryotes is more gel-like than that of eucaryotes. The cytoplasm of bacterial cells consists of aqueous solution of 3 groups of molecules: macromolecules such as proteins (enzymes), mRNA and tRNA, small molecules (energy sources, precursors of macromolecules, metabolites or vitamins; and various inorganic ions and cofactors). The primary structural components found in the cytoplasm are the nucleoid, ribosomes and some type of inclusion.

Nucleoid is irregularly shaped region which contains the chromosome, usually 1 per cell. The nucleoid is not membrane-bound. The chromosomes are usually a closed circular, double-stranded DNA molecule looped and coiled extensively. Other than the main chromosome bacteria also have usually small, closed circular DNA molecules which exist and replicate independently of chromosome. The plasmids are found in the cytoplasm but not in the nucleoid and have relatively few genes present.

Ribosomes are complex structures consisting of protein and rRNA. These are the sites of protein synthesis. In prokaryotic cells they are smaller (prokaryotic ribosomes $\Rightarrow$ 70S, eucaryotic ribosomes $\Rightarrow$ 80S).

The cytoplasmic matrix contains some storage materials and inclusions which can be organic or inorganic: Polyphosphate (Polymetaphosphate) like Babes-Ernst bodies in *Corynebacterium*, Glycogen

granules, Sulfur granules, Lipid droplets, Cyanophycin, (nitrogen polymer in cyanobacteria), Magnetite (Fe_3O_4), Gas vesicles (vacuoles) and Metachromatic granules (Phosphate crystals or volutin granules).

Bacterial endospore: Bacterial spores are dormant forms of bacteria which are thick walled and highly retractile. All spore forming bacteria are gram positive and it depends on the ability to form thick wall. Spore formation is by bacilli and few cocci and common among aerobes or obligate anaerobes. Spores are highly resistant to environmental stresses such as heating, cold, irradiation, strong acids, disinfectants, drying/desiccation etc. In nutritional deprivation or when the supply of C, N & P is limited, certain bacteria develop endospores. Exposure to suboptimal temperature can also give rise to endospores. Endospores are cryptobiotic (no metabolism), showing no sign of life. Their dehydrated state and calcium dipicolinate gives them resistance against heat. They are moved by air or other mediums to a fresh environment. Sporogenesis or Sporulation (Spore Formation) needs 6 to 8 hours and occur through different stages from pre-sporulation phase to lysis of cell and liberation of mature spore.

Endospores can be localized as Central spore (*C. perfringens*), Subterminal spore (*C. septicum*, *novyi*, *histolyticum*, *C. difficile*, *C. botulinum*) or Terminal spore (*C. tetani*).

The endospore have certain structures: exosporium is the first and is made of glycoprotein & Lipoproteins. Spore membrane (wall) is the keratin-like protein consisting sulphur containing amino-acids with disulphide bonds (impermeable, protects spore from chemicals), the inner and outer coats (form the thickest spore layer, made of modified peptidoglycan, susceptible to lysozyme) and Core/cortex (is a cell wall structure similar to that in the vegetative cell wall). The surrounding mother cell is called sporangium.

Exospores (conidia) is an extracellular spore expressed by actinomycetales & fungi. Just like endospore they are cryptobiotic but unlike endospore they are not resistant to heat and disinfectants.

Virus literally means fluid poison, this name may be because they are acellular (no cytoplasm, no organelle) obligate intracellular parasites (outside a host cell they are inert, no enzyme or any other activity). Viruses can't be cultivated on artificial media but can be cultivated on tissue culture, embryonated egg or laboratory animals. No ribosomes or any means of protein synthesis and no ATP generating system in viruses. Since they carry out no metabolism on their own they must replicate using the host cell's metabolic machinery. Viruses are the smallest infectious agents known to infect animals (humans, insects), plants and even other microorganisms like bacteria, fungi and protozoa. But their host range is generally narrow. Every intact virus have at least 2 parts: nucleic acid (DNA or RNA, never both) and capsid (protein coat), together forming nucleocapsid. So every virus will have at least a Nucleocapsid. Capsid is made from protein and protects viral genome from host endonuclease. Capsid is also resistant to drying, heat, detergents, acids and Proteases. The building blocks of the capsid are capsomers.

Some of the viruses will also have an envelope and we call them enveloped viruses. Those without envelope are naked viruses. The envelope is made from double layer of phospholipid. It allows attachment of virus to a new host, facilitating entry to the cell. The protein portion of the envelope is of

viral origin but the lipid portion is of host origin. Unlike capsid envelope is sensitive to drying, heat, detergents and acid.

Except for pox viruses the other viruses can't be seen in light microscopy and needs electron microscopy (ranges from 15 to 3000 nano meter). Their shape can be rod like, spherical, filament (like ebola) or other shapes as well. Viral symmetry can be icosahedral (polyhedral/cubical), helical or complex.

- Icosahedral symmetry show spherical with crystal like appearance (Adenoviruses, HIV, poliovirus, Hepatitis viruses, Yellow fever virus and human papilloma virus).
- Helical symmetry is for the rod shaped viruses where the capsomers are arranged in ribbon (helical) like pattern around the nucleic acid. All helical human viruses are enveloped (Influenza virus).
- Complex symmetry: complicated structures, brick shaped with ridges on the external surface. This type of virus composed of several parts, each with separate shapes and symmetries (pox viruses).

The classification of viruses depends on morphology, nucleic acid type, taxonomy and the type of disease they cause. But the morphologic and nucleic acid type are the best choices.

Based on morphology of the capsid (shape of the coat) we can have many classifications:

- Helical viruses: the protein capsid appears in a coiled pattern. They are non-enveloped which are arranged helically around the virus genome. Examples: Tobacco mosaic virus, Rabies virus
- Polyhedral viruses: nucleic acid surrounded by a hollow protein, multi-sided capsid. The capsid of most polyhedral viruses is in the shape of a regular polyhedron with 20 triangular faces and 12 corners. Examples: Adenoviruses, Polio viruses
- Enveloped virus: roughly spherical but variable in shape. When helical or polyhedral viruses are enclosed by envelopes, they are called enveloped helical and enveloped polyhedral viruses. Examples: Influenza virus, Herpes simplex virus. So based on the outer covering virus can be enveloped or non-enveloped (naked).
- Complex Virus: has neither helical nor polyhedral forms and is combination of both these forms is formerly called complex. Examples: Bacteriophage, Poxvirus

Based on baltimore classification grouped based on their genome (nucleic acid) and the type of replication there are seven types of viruses: dsDNA viruses, ssDNA viruses, dsRNA viruses, +ve sense ssRNA viruses (codes directly for protein), -ve sense ssRNA viruses, RNA reverse transcribing viruses and DNA reverse transcribing viruses.

The classical system of classification is rather simple as it is based on the nucleic acid genome only: DNA viruses and RNA viruses.

- DNA virus families (Parvoviridae, Herpesviridae, Polyomaviridae, Hepadnaviridae, Papillomaviridae, Adenoviridae and Poxviridae)

- RNA virus families (Picornaviridae, Reoviridae, Bornaviruses, Flaviviridae, Filoviridae, Togaviridae, Arenaviridae, Coronaviridae, Bunyaviridae, Caliciviridae, Orthomyxoviridae, Retroviridae, Paramyxoviridae, Astroviridae and Rhabdoviridae)

A fact that should not be forgotten is that all animal RNA viruses code for a polymerase. But the polymerase can be of 3 types: RNA-dependent RNA polymerase (convert RNA to another RNA), RNA-dependent DNA polymerase (also known as reverse transcriptase, converts an RNA to a DNA) or DNA-dependent RNA polymerase (converts the hosts DNA in to an RNA).

Viruses replicate in a lytic or else a lysogenic method. In lytic infection, viruses are released from the cell by lysis while the replication cycle usually results in death and lysis of host cell. But in lysogenic replication the viruses integrate their genomes into that of the host cell, and may be stably inherited through several generations before returning to lytic infection. In the lysogenic, infected host cells grow and reproduce normally for generations before they lyse (for example temperate phages).

The lytic replication have several stages from attachment to the target cell all the way up to the lysis. This stages can be summarized as:

Attachment (adsorption) → Penetration (injection) of virus or its nucleic acid → Early replication steps (host cell biosynthetic machinery altered, virus-specific enzymes made) → Replication of virus nucleic acid → Synthesis of proteins (structural subunits of virus coat) → Assembly structural subunits (and membrane components in enveloped viruses), packaging of nucleic acid → Release of mature viruses (with lysis)

Fungi

About 100,000 species are discovered, most are useful and about 100 are pathogenic to human (none of the pathogenic fungi are obligate intracellular). Fungi are heterotrophs that feed by absorption and unlike bacteria they are eukaryotic. Most fungi are multicellular but yeast is in a unicellular form.

The natural habitat of most fungi is the environment which can present good amounts moisture and organic materials which favors their growth. Very few are normal flora in humans (for example *Candida albicans* & *Malassezia furfur*). Most fungi are obligate aerobes (most molds), some are facultative anaerobes (yeasts) but no fungi can be obligate anaerobes.

Most fungi have cell walls made of chitin (composed of long chain of n-acetyleglucosamine) and other polysaccharides glucans (like manans, chitosan and galactans). Fungi also contain networks of branched hyphae known as mycelia, which is an adaptation for absorption. Unlike the major sterol in human cell membranes (cholesterol), the major sterols in the fungi cell membrane are ergosterol & zymosterol. The antifungal drugs Amphotericin B & Azole group act on the ergosterol.

Another structural information about fungi is the septate and aseptate forms, which are with and without cross walls, respectively. When septate they will have hyphae divided into cells by septa, with pores allowing cell-to-cell movement of materials.

Different fungi exist in different forms as yeasts (single, round to oval cells dividing by budding) or the subunits of molds & mushrooms.

Reproduction in fungi depending on the species can be sexual or asexual. The sexual reproduction is in two ways: Cell fusion (plasmogamy) or nuclear fusion (karyogamy). Asexually, fungi can grow as a mold (at times on foods) and as yeast (reproduce by simple cell division on moist environment). Because most molds and yeasts have no known sexual stage they were known as deuteromycetes (imperfect fungi).

Fungi exhibit different life style: and can be decomposers, parasites on humans/other animals/plants or they could also establish mutualistic relations with plants (example can be lichens).

NORMAL MICROBIAL FLORA(microbiota)--MF

- Means Kind of organisms in various parts of human body
- Can be resident or transient flora

Properties of normal flora

- acquired rapidly during & after birth
- reflects age of person
- changes continuously throughout life
- reflects- nutrition, genetics, environment and sex

Advantages of the normal flora

- colonization resistance = bacterial antagonism
- bacterium-bacterium interaction
 - competition
 - nutrients
 - mucosal attachment sites
 - accumulation of volatile fatty acids
 - release of bacteriocins
 - alterations in pH or oxidation-reduction potential
- maintain normal microbial flora
- antibiotics administration
- Immunity
 - stimulates immune system
- Nutrition-vitamin K, B12, B(enteric and lactic acid bacteria)
- **Detrimental effects of MF**-bacterial synergism and opportunistic pathogens(**E.coli from gut to Urinary Tract and *Klebsiella pneumoniae* from gut to respiratory**)
- **Kind and number of MF is affected by-** oxygen, receptor sites for attachment, pH, availability of nutrients, influence exerted by other microbes, immune status
- **Normal Flora of the Skin**
 - Micrococci, Diphtheroids, *Propionibacterium*(*acne*-so not present in children)
 - Sweating and washing don't affect it.

- **soap or disinfectant** (hexachlorophene) **diminish microbial population** but normal flora rapidly replenished from **sebaceous & sweat glands**
- **Normal Flora of the conjunctiva**
 - Staphylococcus epidermidis, Corynebacterium sp., Propionibacterium acnes, Staphylococcus aureus, Viridans streptococci, Neisseria sp., Haemophilus influenzae
- **Normal Flora of the respiratory tract**
 - Staphylococcus epidermidis, Corynebacteria, Staphylococcus aureus, Neisseria sp, Haemophilus sp, Streptococcus pneumoniae
- **Normal Flora of the gastrointestinal tract**
 - **Children**-at birth-intestine is sterile
 - **In breast-feed**-Bifidobacteria->90% of intestinal MF
 - **In bottle-fed infants**-Bifidobacteria are joined by *Enterococci, Bacteroides, Staphylococci, Lactobacilli and Clostridia*
- **upper GIT of adult humans**
 - mainly acid-tolerant lactobacilli present *e.g. Helicobacter pylori*
- **in small intestine**
 - Lactobacilli, *Enterococcus faecalis*, Coliforms and Bacteroides

BACTERIAL GROWTH AND NUTRITION

- **Generation time:** Time required for a population of bacteria to double in number *Most pathogenic bacteria: 30-60 min (20 min)*

Phases of Microbial Growth

I. Lag phase

- Period of adjustment/adaptation to the environment
- Enzymes and intermediates are synthesized
- Growth rate = 0

II. Exponential/logarithmic phase

- Growth rate is constant = 2^n (doubling in numbers at regular intervals)
- Exponential growth is balanced growth, cell components are synthesized at constant rates relative to another

III. Stationary phase

- Balance between cell division and cell death (nutrient depletion + accumulation of toxic waste products)
- Growth rate = 0

IV. Death phase

- Cell viability lost, growth rate negative ((nutrient depletion + accumulation of toxic waste products)
- Number of viable cells decreases geometrically (exponentially), essentially (the reverse of growth during the log phase)
- Growth rate –ve

Culture

- Process of cultivation/growing of M.O's by providing appropriate nutrients and environmental conditions

Culture media

- Nutrition composed of different elements to grow, transport and store of M.O's
- **Forms of culture media**
 - Solid media-Nutrient + 1-2% agar
 - Colony seen here.
 - Liquid media: Contain only nutrients, no agar
 - Microbial growth characterized by observing turbidity
- **Composition of media**
 - **Defined/synthetic media:** Contains ingredients/ nutrients, their compositions/components are known (chemically defined)
 - **Complex media:** Media that contain ingredients/nutrients of unknown chemical composition Contain peptones, meat and yeast extracts serve as sources of carbon, energy, nitrogen, A,A's, vitamins and minerals
- **Types of culture media based on function**
 - **General purpose media:** Support the growth of many microorganisms e.g. nutrient broth
 - **Enriched media:** Enriched with blood/serum/vitamins to encourage the growth fastidious M.O's e.g. blood and chocolate agar
 - **Selective media:** Contain antimicrobial agents/ inhibitory substances to favor the growth of particular microorganisms e.g. Thayer-Martin media
 - **Differential media:** distinguishes between different groups of bacteriae.g. MacConkey agar (lactose fermenter vs. non lactose fermenter)
 - **Transport media:** Maintain the viability of the bacteria in the specimen until processed/cultured e.g. Stuart media

BACTERIAL NUTRITION

- Most pathogenic bacteria are chemoheterotrophs-**meaning use preformed organic molecule and obtain energy from its oxidation.** (organic compounds as carbon and energy sources)

UPTAKE OF NUTRIENTS BY THE BACTERIAL CELL

- **Passive/Simple diffusion-no energy needed**

- Small molecules such as H₂O, O₂ and CO₂ often moves across membranes by passive diffusion
- **Facilitated diffusion**
 - Transport of solutes in and out of the cell as regulated by a carrier membrane protein (permeases)
- **Group translocation**
 - Chemical modification of a molecule needed and energy is required
- **Active transport**
 - Transport of molecules to higher concentrations or against concentration gradient with the use of energy (without modification of molecules) Membrane carrier proteins needed
- **Iron uptake:** Iron binding proteins (siderophore)
 - able to form complex with ferric iron (Fe³⁺) and reduced to the ferrous form (Fe²⁺)

Sterilization and disinfection

- **Sterilization**_use of **physical and/or chemical** procedures to **completely eliminate** or destroy **all forms of microbial life** (including resistant bacterial spores)
- **Disinfection** – process of elimination microorganism with the exception of spores
- **Antiseptic** -**disinfectant applied directly to body surfaces** /are not ingestible.
 - used for hand hygiene-e.g.**chlorhexidine, iodophors,alcohol-based handrub**
 - Skin, cervical, or vaginal preparation before a clinical procedure
- **Common antiseptics**
 - **Alcohol**-(60-90% ethanol or isopropanol)
 - Intermediate/low level of disinfectant but is **flammable**
 - Fast acting, non-toxic, **non-staining**, easily evaporates
 - Cannot penetrate protein rich materials
 - skin antiseptics
 - **Iodophors**- Povidine in alcohol:for**Pre-surgical Skin Prep**and as antiseptics.
 - **Hydrogen peroxide**-aka chemical sterilant
 - Antiseptics, cleaning contact lenses, Ventilators,

Disinfectants (Chemical)-kill all microbes except large spores. e.g.

- **Glutaraldehyde-containing products**-used for **laparoscopes; endoscopes**
- **Chlorine**-either from bleach powder or house hold bleach or tablets.
 - high level disinfection and decontamination of instruments
- **Peracetic acid,Formaldehyde,quats,phenol**

Sterilization

- **Thermal**
 - **Dry heat**-Dry oven or hot air sterilizer-recommended exposure: 140°C/3h ; 150°C/2.5h; 160°C/2h; 170 °C/1h
 - incineration,Laboratory Bunsen burner**

- **Moist heat**-Steam sterilization –saturated heat under pressure
 - AutoclaveAdvantages: Inexpensive, nontoxic and very reliable
 - used for any material that can withstand heat and **steam**

Introduction to Pharmacology

☞ Basic definitions:

- **Pharmacology** is the body of knowledge concerned with the action of chemicals on biologic systems.
- **Medical pharmacology** is the area of pharmacology concerned with the use of chemicals in the prevention, diagnosis, and treatment of disease, especially in humans.
- **Toxicology** is the area of pharmacology concerned with the undesirable effects of chemicals on biologic systems.
- **Pharmacokinetics** describes the effects of the body on drugs, eg, absorption, excretion, etc.
- **Pharmacodynamics** denotes the actions of the drug on the body, such as mechanism of action and therapeutic and toxic effects.
- *Pharmacogenetics – the branch of genetics that studies the genetically determined variation in response to drugs in human or laboratory organisms. It is very important to identify therapeutic failures and unanticipated toxicity.*

Drugs are substances that act on biologic systems at the chemical (molecular) level and alter their functions.

- **Physical nature** (at room temperature): solid (aspirin, atropine), liquid -- nicotine, ethanol or gaseous (nitrous oxide)
- **Drug – Receptor Interaction: agonist** (activation of receptors) or **antagonist** (inhibition)

A few drugs are enzymes themselves (eg, thrombolytic enzymes that dissolve blood clots; pegloticase, which metabolizes uric acid). These drugs do not act on endogenous receptors but on endogenous substrate molecules.

Inert binding sites: are other non-regulatory molecules in the body that the drug binds to, and obviously no discernible effect. In the plasma buffer the concentration of a drug because bound drug does not contribute directly to the concentration gradient that drives diffusion.

- **Size:** MW 7 (lithium) to MW 50,000 (thrombolytic enzymes, other proteins). Alteplase [t-PA], a clot-dissolving enzyme is a protein of MW 59,050. Most drugs however range between MW 100 and 1000.
- **Drug Reactivity and Drug-Receptor Bonds:** covalent (very strong, irreversible action), electrostatic (weaker, highly selective short-acting drugs) or hydrophobic (very weak)
- **Source**
 - **Natural:** plants (Morphine, Digoxin), animals (Insulin), soil (Iron) and even microorganisms (Antibiotics)

- **Synthetic** - Aspirin
- **Semisynthetic** - Codiene

Pharmacokinetics

Major processes involved are absorption, distribution, metabolism, and elimination. It will enable us to calculate loading and maintenance doses.

Effective drug concentration: the concentration of a drug at the receptor site. In patients, more readily measured in the blood (except for topically applied agents). The **plasma concentration** is a function of the rate of input of the drug (by absorption) into the plasma, the rate of distribution, and the rate of elimination. If the rate of input is known, the remaining processes are well described by 2 primary parameters: apparent volume of distribution (Vd) and clearance (CL). These parameters are unique for a particular drug and a particular patient but have average values in large populations that can be used to predict drug concentrations.

Volume of distribution: relates the amount of drug in the body to the plasma concentration according to the following equation:

$$V_d = \frac{\text{Amount of drug in the body}}{\text{Plasma drug concentration}}$$

A drug that is completely retained in the plasma compartment will have a Vd equal to the plasma volume (about 4% of body weight). The Vd of drugs that are normally bound to plasma proteins such as albumin can be altered by liver disease (through reduced protein synthesis) and kidney disease (through urinary protein loss).

If a drug is avidly bound in peripheral tissues, the drug's concentration in plasma may drop to very low values even though the total amount in the body is large. As a result, the Vd may greatly exceed the total physical volume of the body.

CLEARANCE (CL): relates the rate of elimination to the plasma concentration.

$$CL = \frac{\text{Rate of elimination of drug}}{\text{Plasma drug concentration}}$$

For a drug eliminated with first-order kinetics, clearance is a constant. But for drugs eliminated with zero-order kinetics, clearance is not constant. Clearance depends on the drug, blood flow, and the condition of the organs of elimination in the patient. Disease, age, and other variables usually alter the clearance of a drug much more than they alter its Vd.

The clearance of a particular drug by an individual organ is equivalent to the extraction capability of that organ for that drug times the rate of delivery of drug to the organ.

The clearance of a drug that is very effectively extracted by an organ is often flow-limited. Other conditions like cardiac disease, or other drugs that change blood flow may have more dramatic effects on clearance than disease of the organ of elimination.

Half-life: time required for serum plasma concentrations of drug to decrease by one-half (50%). This parameter is completely determined by V_d and CL . Like clearance, half-life is a constant for drugs that follow first-order kinetics. Half-life is clinically important as short half-life → drug is removed efficiently.

$$t_{1/2} = \frac{0.693 \times V_d}{CL}$$

The half-life determines the rate at which blood concentration rises during a constant infusion and falls after administration is stopped.

BIOAVAILABILITY: the fraction (F) of the administered dose that reaches the systemic circulation. Defined as unity (or 100%) in the case of intravenous administration. After administration by other routes, bioavailability is generally reduced by incomplete absorption (and in the intestine, expulsion of drug by intestinal transporters), first-pass metabolism, and any distribution into other tissues that occurs before the drug enters the systemic circulation. Even for drugs with equal bio-availabilities, entry into the systemic circulation occurs over varying periods of time, depending on the drug formulation and other factors.

To account for such factors, the concentration appearing in the plasma is integrated over time to obtain an integrated total area under the plasma concentration curve (AUC). **Area under the curve (AUC) - the graphic area under a plot of drug concentration versus time after a single dose or during a single dosing interval.**

EXTRACTION: removal of a drug by an organ can be specified as the extraction ratio, that is, the fraction or percentage of the drug removed from the perfusing blood during its passage through the organ. After steady-state concentration in plasma has been achieved, the extraction ratio is one measure of the elimination of the drug by that organ. Drugs that have a high hepatic extraction ratio have a large first-pass effect; the bioavailability of these drugs after oral administration is low.

DOSAGE REGIMENS: a plan for drug administration over a period of time. An optimal dosage regimen results in the achievement of therapeutic levels of the drug in the blood without exceeding the minimum toxic concentration. To maintain the plasma concentration within a specified range over long periods of therapy, a schedule of maintenance doses is used. If it is necessary to achieve the target plasma level rapidly, a loading dose is used to “load” the V_d with the drug. Ideally, the dosing plan is based on knowledge of both the minimum therapeutic and minimum toxic concentrations for the drug, as well as its clearance and V_d .

THERAPEUTIC INDEX – Narrow or Wide

Pharmacokinetics mainly deals with: Volume of distribution & Clearance (Half-life), Bioavailability (First pass effect) and Dosing (Maintenance and Loading).

Route of administration: **depends on** - physicochemical properties of the drug, rate and extent of absorption, site and onset of desired action, first pass metabolism, dosage form (accuracy of dose) and status of the disease.

- Topical route: skin or mucus membrane
- Inhalation (lung) –
 - Depends on partial pressure of gases (increased PP – important for targeting)
 - prevents first pass metabolism
 - i.e. **First-pass effect, presystemic elimination** -- The elimination of drug that occurs after administration but before it enters the systemic circulation (eg, during passage through the gut wall, portal circulation, or liver for an orally administered drug)
- Enteral: oral, sublingual (nitroglycerin – for severe chest pain caused by increased work load of heart. It decreases the work load of heart by dilating veins) or rectal (for local activation)
- Parenteral
 - Intravenous: Ideal for irritant drugs, In solution form
 - Intramuscular: Absorption depends on the rate of skeletal blood flow, Irritating drugs can be administered
 - Subcutaneous: Slower onset than IM, drugs must be non-irritant
 - Depot therapy – Absorption is slow & constant
 - Pellets inserted below the skin (Testosterone)
 - Retardation of absorption – Vasoconstrictors
 - Others: Intradermal, Intra-arterial (Anticancer drugs target directly), Intracardiac, Intraperitoneal and Intrathecal

Mechanism of Absorption

- ⇒ **Filtration:** Drug passes through the pores (Small molecules & Water solubility)
- ⇒ **Passive Transport:** Simple Diffusion or Facilitated Diffusion
- ⇒ **Active Transport:** **Against** concentration gradient. Requires carrier molecules, Saturated, Selective, Energy Dependent

Factors affecting absorption: Dosage form (Solid/liquid/gas), Solubility of the drug, Particle size of the drug, Ionization, Surface area, Gastric motility, Rate of regional blood flow, Presence of food or other substances (Insulin – before meal) and Disintegration & dissolution time

Ionization

The electrostatic charge of an ionized molecule attracts water dipoles and results in a polar, relatively water-soluble and lipid-insoluble complex. Ionization of drugs may markedly reduce their ability to permeate membranes.

A weak acid is best defined as a neutral molecule that can reversibly dissociate into an anion (a negatively charged molecule) and a proton (a hydrogen ion). For example, aspirin dissociates as follows:



Protonated form of a weak acid and the unprotonated form of a weak base are neutral and more lipid-soluble. **Law of mass action:** these reactions move to the left in an acid environment (low pH, excess protons available) and to the right in an alkaline environment. Weak acidic drugs ionize more in basic medium.

Distribution

Factors affecting distribution: Membrane permeability (cross membranes to site of action), Lipophilicity of drug (lipophilic drugs accumulate in adipose tissue) Ionization at physiological pH (a function of its pKa, drugs that are charged, especially positively charged, tend to bind to serum proteins), Presence of tissue-specific transporters, Differences in regional blood flow, **Pathological states** can alter V_d (Congestive heart failure, Cirrhosis of liver) and Lipid : water partition coefficient of the drug (Number of shift from blood to adipose)

Enhancing elimination by dialysis depends on V_d . If V_d is high, enhanced elimination will probably be fruitless. Drugs with high V_d are not confined to the vascular compartment. Dialysis of the blood, therefore, does not reach the major fraction of the drug because it is hidden away at receptor sites.

Remember: Heparin (Plasma binding + very large), Gentamicin (water soluble), Ethanol (Less water soluble), DDT (highly water soluble), Aspirin in gout increases uric acid level

NB: Low V_d drugs are good candidates for drug monitoring.

- Psychotropic drugs - Binding at sites within the central nervous system. CNS is quite remote from the blood so not suitable subjects for therapeutic drug monitoring.
- Plasma protein bound drugs - Restricted to vascular compartment. Malnutrition = ↓albumin = ↑ free drug. High protein binding means long acting (not metabolised).
- **Reservoir** - Bound drugs are in equilibrium with unbound drug.

Free and bound drugs

Total drug (bound plus free) should be measured and expressed in laboratory result. Measurement of the free fraction is often clinically preferable especially for drugs with high degrees of protein binding such as phenytoin. (i.e. The drug's free fraction is likely to increase in overdose.) Phenytoin is normally 10% free.

Biotransformation: Pro-drug is usually administered, will be later activated.

Factors affecting biotransformation: Race (CYP2C9; warfarin (bleeding) , phenytoin (ataxia) also fast and slow isoniazid acetylators), Age (reduced in aged patients & children), Sex (women slower ethanol metabolizers), Clinical or physiological condition, Other drug administration (induction or inhibition) and Food (grapefruit juice --CYP3A).

Drug excretion: Via Kidneys (GFR, Tubular reabsorption, Tubular secretion), GIT (through feces, don't forget enterohepatic circulation), Lungs, Saliva, Skin and Breast.

Polar drugs pass into the urine. But the nonpolar drugs are reabsorbed in the renal tubules (persist weeks or even months).

- **First order:** when the concentration of the drug in plasma declines with time and the rate of the elimination also slows. **This means rate is a function of the concentration.**
 - **This is characteristic of a toxin that is primarily eliminated by the renal route.**
- **Zero-order:** elimination is characterized by the plot of concentration vs. time being a straight line, meaning **rate of elimination is not influenced by concentration.**
 - The toxin or drug is eliminated at a constant rate irrespective of whether there are small or large amounts of it (Characteristic of hepatic elimination for nonpolar drugs).
 - **Rate of zero-order elimination** depends on the **availability of hepatic enzymes** and these are limited. At enzyme saturation, which occurs usually at a low drug concentration, the reaction rate is maximal.
 - The concentration of metabolizing enzyme, not the concentration of toxin, is the major factor in setting the rate of elimination (Acetaminophen).
- **Michaelis-Menten kinetics** (mixture of zero- and first-order pharmacokinetics) Ex: Alcohol
 - At a high concentration of drug the rate of elimination is dictated by the concentration of the fully saturated enzyme. At this concentration zero order prevails and elimination is not related to drug concentration; the elimination rate is constant. At a certain drug concentration, the amount of drug or toxin is too low to saturate the metabolizing enzymes. Beyond that point the reaction rate is a function of the concentration of toxin. The rate becomes first order.
- Drug should have two characteristics: Rapid loss in the kidney and Metabolising enzyme is present in large amount.

PHARMACODYNAMICS

Pharmacodynamics is the study of the biochemical and physiological effects of drugs (at treatment, prevention or diagnosis) and their mechanism of action. It is mainly concern on interaction of the drug with receptors, but receptors are not always required to have an effect. The principles of pharmacodynamics apply to all biologic systems, from isolated receptors in the test tube to patients with specific diseases.

Pharmacodynamics is very important:

- To provide the basis of the rational therapeutic use of a drug, Example: vasodilators for management of hypertension, ischemic heart disease and erectile dysfunction
- To manage poisoning
- For the design of new superior therapeutic agents, Example: me-too drugs, the thalidomide case (used to be taken for the treatment of sleep disorder and nausea, but later used in the treatment of cancer because it affects or inhibits angiogenesis)
- To understand the untoward effect of the drug, Example: hypoglycemia from antidiabetic agents

Biochemical and physiological change after drug administration mainly involve target site interaction of the drug with macromolecule.

- Drug targets
 - Receptor mediates (specific drug targets): receptors can be **proteins**, **membrane sterols** (for fungi infection ergosterol - create pore at fungi membrane and facilitate influx of electrolyte) or even **nucleic acids** (for cancer Rx, decreases replication by inhibiting DNA replication through alkalinizing agents, prevent DNA splicing)
 - Non receptor mediated (non-specific) drug targets
 - Water - drugs affecting water movement
 - ☞ Osmolar agents: Increase osmolar gradient
 - ☞ Diuretic agents: Increase urination (taken IV → at tubular area → increases osmolarity gradient → facilitate water movement to tubular area → increased urination)
 - H and OH ion: Antacids ($\text{Mg}(\text{OH})_2$, $\text{Al}(\text{OH})_3$)
 - Metal ions: during metal ion in-toxication (Mg, Fe...) → Chelating agents are given (form complex with the metal ions so that the killing agents are absorbed less)

NB - Most of the interactions result in receptor mediated response. Most drug receptors are chemically proteins or derivative of proteins (sometimes nucleic acid, cell membrane...). Most receptors are specific and saturable.

Classification of receptors

- ~ Protein receptors
 - Regulatory receptors ----Regulate body activity – nervous system, Endocrine system (via hormonal receptor), ...
 - Enzymes ---- Involved in synthesis or degradation (Acetylcholinesterase)
 - Ion channels ----- Na/K ATPase (Depolarization/ Hyperpolarization)
 - Structural protein ----- Micro tubule (targeted to influence cell division)
 - Transport protein / uptake channel
- ~ Location of drug receptors
 - Cell surface receptors

- Intracellular receptors: drug should be lipophilic to cross cell membrane and enter the cell so a delayed response. But the effect can be maintained (persistence).
- ~ Receptor can be in
 - The host ----- hypertension
 - Disease causing microorganism ----- Antimicrobial (i.e. Selectivity is mandatory)

NB – Receptors are NOT ACCEPTORS. ACCEPTORS are Plasma protein. Though acceptors have a binding site there is NO EFFECT on body and it ONLY affects drug distribution. But in case of receptors the binding of the drug will have an effect because of interactions.

Type of receptor binding: influences the response (i.e. Most interactions involve bonds of multiple types). Reversible binding are hydrogen bond and ionic bond. Slowly reversible or irreversible are mainly covalent bond and some non-covalent bond. In case of the irreversible the bond is strong, so the effect of the drug may persist, though the drug is not found in the plasma.

Example: Phenoxy benzamine – an alpha blocker (irreversible and through non-competitive manner, may result in hypertension by pheochromocytoma by increasing level of epinephrine though the plasma concentration is zero (i.e only the concentration of drug @ the target matters. But still plasma concentration indirectly indicate the target organ content as there is no other reliable means of measuring the organ content directly).

Ligand

Drugs or toxic chemicals or endogenous compounds binding to receptors. Ligands could be:

- Agonist – produce similar effect as that of endogenous ligand
- Partial agonist – agonist producing partial effect. In the presence of full agonist, it may act as antagonist
- Antagonist – Oppose endogenous ligand but does not cause response by themselves
 - Competitive – reversible or weak binding (eg: beta receptor agonist and antagonist)
 - Noncompetitive – Non reversible or strong binding Vs allosteric binding (eg: phenoxybenzamine with epinephrine on alpha 1 adrenergic receptor)
- Inverse agonist – stabilizes the receptor in its inactive conformation
 - An agent which binds to the same receptor binding site as an agonist for that receptor and reverses constitutive activity of receptors
 - A prerequisite for an inverse agonist response is that the receptor must have a constitutive (also known as intrinsic or basal) level activity in the absence of any ligand

Terminology

- ✓ Affinity is drug attraction for the receptor: High affinity (low concentration bind), Low affinity (high concentration bind) or No affinity (does not bind)
- ✓ Intrinsic activity – relative ability of a drug-receptor-complex to produce a maximum functional response (EFFICACY)

- ✓ Potency – the dose required to produce a given effect, commonly expressed by ED_{50}
- ✓ ED_{50} – the concentration of a drug that is effective in 50% of the participants/patients OR the concentration of the drug that cause 50% of the maximum effect of a drug
- ✓ Onset of action – time between drug delivery and start of an effect
- ✓ Duration of action – time gap of a drug that maintain the effect

Maximum effect: effect seen when all receptors are occupied or saturated. It is rarely achieved as there are spare receptors (meaning all receptors won't be occupied). Occupancy principle: Partial agonist cannot reach maximum effect Without full occupancy but Full agonist can reach maximum effect without full occupancy.

2 state receptor model: Two conformational state:

- Receptors have two forms: active [R_a] and inactive [R_i] form (their balance determines the presence of basal activity): Full agonist go to active state of receptor while the partial agonist go to the inactive state of receptor. Antagonist have an equal affinity for both forms.
- Constitutive receptor activation (activation even with no ligand present): Benzodiazepines, Cannabinoids and Serotonin.

Signal transduction pathways (STP)

Signal transduction may either involve protein modification or phosphorylation. Receptor drug interaction may involve signal transduction through secondary messengers. (i.e. Secondary messengers are intermediary molecules between the ligand-binding domain and effector domain)

Receptor type based on STP

- ⇒ Ligand gated ion channel (ionotropic receptors): effect in milliseconds
 - Binding of ligand → Open / Close ion channel → Depolarization / Hyperpolarization
 - Eg: Nicotine receptor, GABA A receptor, 5-HT₃ receptor
- ⇒ G-protein coupled receptor (metabotropic receptors or 7-TM spanning (heptahelical) receptors): effect in seconds, involve secondary messengers.
 - Binding of ligand → G-protein released → activation of ion channel
- ⇒ Enzyme linked receptor: effect in hours
 - Could be protein kinase linked or Guanylyl cyclase linked
 - Eg: Insulin, growth factors, cytokins Vs atrial natriuretic factor
- ⇒ Nuclear receptor: effect in hours
 - Inside a cell (in the cytosol or nucleus) so drug has to be lipophilic. It have delayed and prolonged effect as it requires DNA replication and protein synthesis as a response
 - Eg: Thyroid receptor, Steroid receptor (Estrogen receptor)

G protein coupled receptors involve secondary messengers

- Most common drug receptor: muscarinic receptors, adrenoceptors, dopamine receptors, 5-HT receptors, opiate receptors, purine receptors

- Has 7 trans-membrane alpha helices
 - o Extracellular N-terminal domain
 - o Intracellular C-terminal
 - o 3 subunits (alpha, beta, gamma)
- ⇒ Type of effector targets
 - o Enzymes
 - Adenylyl cyclase (G_s) → cAMP [When activated → contraction by increasing phosphorylation, responsible for heart failure]
 - Guanylyl cyclase (G_i) → cGMP [When activated → Relaxation by increasing dephosphorylation, responsible for erectile dysfunction]
 - Phospholipase (G_q) → IP3 and DAG
 - o Ion channel
 - Calcium, potassium
 - o Rho A/ Rho kinase – a system that controls the activity of many signaling pathways controlling cell growth and proliferation, smooth muscle contraction ...

Control of receptor expression

Short term regulation /Desensitization/ ---the second main processes involved: Receptor phosphorylation and Receptor internalization (endocytosis). Long term regulation – through increasing or decreasing receptor expression. See the following terms:

- ★ Tachyphylaxis – diminish in effect within a minute
- ★ Tolerance – gradual decrease in response to drugs, taking days or weeks to develop
- ★ Drug resistance – loss of effectiveness of antimicrobial or antitumor drugs

Mechanism for the above three phenomena

- Change in or loss of receptor
- Mediator exhaustion --- amphetamine
- Increased / Decreased metabolic degradation of the drugs ---- phenol barbiturates (self-induced potentiation of enzymes metabolize them) Vs nitrovasodilators (prodrug, where decreased metabolism could result in tolerance; Normally NO → Vasodilators)
- Physiological adaptation --- thiazide diuretics
- Active extrusion of drug from cells --- esp in cancer therapy
- Microorganisms may decrease the influx of antimicrobials → RESISTANCE or may increase efflux by expressing reverse transcriptase.

Dose response relationship - the degree of effect produced by a drug is generally a function of the amount administered – expressed by dose response curve → a sigmoid curve. Concentration of drug in plasma is directly proportional with concentration of drug in receptor and response.

- From the curve: E_{max} , EC_{50} or ED_{50} (which indicate potency), LD_{50} and potency
- ★ Two types of dose response curve

- Graded dose response – response continuously increase as the administered dose continuously increase, more common as it involve single subject.
 - Possible to calculate E_{max}, ED₅₀, Therapeutic index (TI... safe if wide) (steeper slope = Narrow toxicity)
 - TI (therapeutic index) = min toxic conc./min effective conc. = LD₅₀/ED₅₀
 - Use of antagonist with agonist shifts the curve to the right
 - Greater potency is not the same as therapeutic superiority
- Quantal dose response – show the effect of different dose on response among all (many) individuals taking the drug. Follow none or all for dose administration and is a cumulative plot of the normal frequency distribution.

Drug interaction

Drugs could have an interaction with agents that are administered to the body concomitantly. Drug interaction could be significantly important or harmful.

Type of interaction

- ★ Drug-drug interaction: Example - antacids decreases the absorption of ketoconazole because acidic environment increases ketoconazole absorption.
- ★ Drug-food interaction
 - Absorption of **efavirenz** (usually given for HIV pts.) is increased (600X) with the presence of **fatty foods**
 - **Grape fruit** juice are CYP3A4 enzyme inhibitor, decreasing the metabolism of drugs by this same enzyme like **warfarin** leading to toxicity.
 - **Dairy products** affect antibiotic (**tetracycline**) absorption by forming complex
- ★ Drug-beverage interaction
 - Both **benzodiazepine** and **alcohol** CNS depressant effect (additive sedative effect)
 - **Wine** have tyramine which have a sympathetic effect and increase the synthesis/release of NE, **MAO inhibitors** inhibit the breakdown of NE → excess NE → hypertensive crises
- ★ Drug-beverage interaction
 - St. John's is liver microsomal enzyme inducer increase metabolism of drugs like warfarin → loss of therapeutic outcome

Drug interaction based on its level, classified as pharmacokinetic and pharmacodynamic interaction.

Pharmacokinetic level interaction

1. Absorption

Reason for interaction – pH alteration, complexation, adsorption, GI motility, GI blood flow, eradicating normal flora. i.e. Best drug interaction is when administered orally.

- FPM (first pass metabolism) – alteration of drug before reaching the systemic circulation (sometimes by normal flora)

- Adsorption – accumulation of molecule of a gas form a thin film on the surface of a solid
- Chelation – removing heavy metal from blood stream by a chelate

Ergotamine is better absorbed from small intestine hence caffeine increases motility decreasing emptying time = facilitate ergotamine absorption from SI (high SA and alkaline media)

Cholestyramine is large macromolecule with a potential to adsorb other agents → FeSO₄ adsorbed by Cholestyramine → decreased absorption

TTC + antacid: tetracycline need acidic environment for better absorption, antacids contain heavy metals and TTC is a chelating agent (decreased absorption because of complexation) and since most antacids are gely and massy adsorption may take place decreasing absorption.

2. Distribution

Main factors for interaction are: high plasma binding and low excreting rate. So drug displacement → increases plasma level → toxicity. Eg, Warfarin + phenylbutazone → warfarine displacement → bleeding

3. Biotransformation

A drug may induce or inhibit the liver microsomal enzyme resulting in drug interaction.

- Enzyme inducer – Phenobarbitol, carbamazepine, phenytoin, rifampin, NVP & EFV, gresofulvin
- Enzyme inhibitor – Cimitidine, ketoconazole, erythromycin, isonazide, CAF, omeprazole, grape fruit juice

Cimitidine + warfarin = bleeding, Phenobarbitol + OCP = pregnancy

First inhibition at equal affinity as induction requires longer time. Rarely, induction may come first if it have higher affinity.

4. Excretion

Through the renal system, mostly at #2 level

- Tubular secretion – since there is carrier case

Eg: Penicillin have short duration of action → probencide increase duration by decreasing excretion

- Tubular reabsorption – since there is pH based ionization of drug and further degree of reabsorption

Eg: weak acid drug + bicarbonate --- increase excretion

Pharmacodynamic interaction

- ✓ Agonizing interaction

- Additive/ summation -----1+1=2 , carbachole + metanichole

- Potentiation---- $1+0>1$, caffeine + ergotamine , loop diuretics + cardiac glycosides
- Synergistic----- $1+1>2$, adrenaline + cardiac glycoside
- ✓ Antagonizing
 - Chemical – chelating agent (dimercaprol) + heavy metals (Hg, Au...) , heparine + protamine = no effect as they are alkaline and acidic substance
 - Functional (physiologic) – interaction of two agonist that act independently of each other but happen to cause opposite effect. Acetylcholine + epinephrine (PSN Vs SN)
 - Competitive – competition of agonist and antagonist for the same site in receptor
 - Depending on the bond formed between antagonist and receptor competitive antagonist:
 - Reversible – bond formed is reversible (loose) by antagonists or agonists (concentration dependent). = parallel shift. Variant (increased) ED50 but equal Emax.
 - Irreversible (not equilibrium) – If bond is covalent, as the concentration of antagonist increase; the slope of agonist curve decreases, the maximum response decreases, No change of ED50. i.e When the amount of antagonist is adequate no amount of agonist can produce any effect.
 - Noncompetitive – involve two binding site (Active site for agonist. Allosteric site for antagonist). Reversible binding, similar dose response curve only difference is specificity.

Factors affecting drug response that may vary among individuals due to: environment, age, sex, diet, weight (SA), genetic factor, tolerance, disease state...

Pharmacogenetics

Fraternal twins Vs identical twins show 75 – 85% of variability in half-life of antipyrine due to genetic polymorphism. It is estimated that genetics (which is stable throughout a person's lifetime) can account for 20-95% of variability in drug disposition and effects.

Polymorphism - natural variation of DNA sequence that have no adverse effect on the individual; and occur in >1% of the general population, due to:

- Single nucleotide polymorphism (SNP) – a change in one base pair of the nucleotide of DNA sequence.
 - Synonymous (silent)
 - Non synonymous (replacement): missense, non-sense: occur at regulatory site (noncoding) or coding
- Insertion or deletion of nucleotide

Harmful effect of drug

Dose matters more!

Harmful effects of drug results from either the drug administered, its metabolite, drug delivery system, or pharmaceutical additives. Side effects are expected adverse effects. Sometimes these side effects

might be beneficiary for the treatment of other underlying disorder. Eg. Corticosteroid increases tension and glycemic level (as side effect) = beneficiary for hypoglycemic patients. ADR accounts for 1/5 of death and serious disability among hospitalized patients. So adverse effect or untoward effect should be minimized or managed for effective drug administration.

- Based on severity: Severe (disability, death), Moderate (need change of therapy and adequate treatment for the ADR) or Mild (need change of therapy).
- Based on onset: Acute (60 min), Subacute (20 hr) or Latent (2 day)

Factors affecting ADR

- **Concomitant infection** (especially in aged population receiving multiple drug therapy/polypharmacy), **Age** (elderly and children are more vulnerable) and **Gender** (on few drugs, eg. Low ethanol metabolism in female)
- Pregnancy or breast feeding (teratogenic effect on the infant or fetus as breast is one of the route of drug excretion. Eg. barbiturates, benzodiazepine, TTC...)
- Comorbid situation (Liver disease and kidney disorder which are essential organs for metabolism and excretion -CAUTION!!!)
- Race, genetic polymorphism: those with poor metabolism → exposed to high dose → TOXICITY. For example: Acetylate metabolizer enzyme on Slow acetylators (administering Isoniazide (30mg) will result in peripheral neuropathy)

Classification

Related to pharmacological action:

- Predictable → extended action of drugs
 - ~ Antiepileptic and anxiolytic → Sedation, Antihypertensive → Hypotension, Insulin → Hypoglycemia, Anticoagulant → Bleeding
- Unpredictable → because of poor selection

Unrelated to pharmacological action:

- ADR at increased dose
 - ~ Paracetamol (block COX enzyme → decreased PG synthesis → Reduces fever) = Hepatotoxicity
 - ~ Aminoglycoside (antibacterial by decreasing protein expression) = Ototoxicity
- Teratogenicity (like tetracycline)
- Under predisposed condition
 - ~ Primaquine induced hemolytic anemia under deficiency of G6PDH
 - ~ Dysarrhythmia from cardio glycoside on hypokalemic patients
- Poor specificity to receptors
 - ~ Antipsychotic or antidepressants block dopamine 2 and also histamine, muscarinic... = untoward effect

Alphabetic classification of types of adverse drug effects

A = Augmented pharmacologic effect

- Common and dose related

B = Bizarre effect

- Uncommon due to hypersensitivity to drug (Anaphylactic reaction on penicillin allergy patients)
- Occur unpredictably and mortality is increased

C = Chronic effect

- During prolonged treatment (Corticosteroid given for cushing syndrome = Inflammation)

D = Delayed effect

- There is large time gap between the time of drug taken and ADR
- Eg. Diethylstilbestrol taken at pregnancy = Vaginal carcinoma to daughter after 20 yr

E = End of treatment effects (withdrawal effects)

- When a chronically taken drug is suddenly withdrawn, mainly for drugs causing addiction

Teratogenicity

Teratogenicity is any structural, functional or behavioral abnormality on the fetus caused by teratogens (chemicals, disease, DRUGS /parent drugs or metabolites of drugs...). Changes are usually irreversible, but not always. May occur at different stages of fetus development:

- Blastocyst (0-16 days) ----- Thalidomide
- Organogenesis (17-60 days) ----- Phenytoin
- Histogenesis and functional maturation (60 days - term) ----- Alcohol, TTC
- Factors: Highly lipid soluble drugs, Unionized, Small MW, Non protein bound, Amount, Placental or fetal circulation

Classification of teratogens

A = No risk on pregnant mother or fetus... SAFE ----- Amoxicillin

B = No risk on animals, No study on pregnant mother... but SAFE ----- Methyl dopa

C = No study/ report With chance of having teratogen ---- Furosemide

D = Teratogenic abnormalities on animals ----- Phenytoin

Thalidamide tragedy given for motion sickness = fetus anomalies

Phenytoin = Craniofacial anomalies

Nealpatone (antiepileptic) = teratogenic effect

Fetal alcohol syndrome (FAS): functional and structural abnormality, poor IQ and craniofacial anomalies

X = Contraindicated ... never given unless it is the only drug to be administered ---- Warfarin, Retinoid

Management of ADR and Drug intoxication

- For nonimmunologic ADR
 - Decrease dose for dose related ADR
 - Switch drug if the drug is used for chronic treatment
- For immunologic ADR: Stop! For hypersensitivity reaction and unpredictable ADR. Managed by:
 - ~ Decreasing the extent of drug in the plasma
 - ~ Prevention of further drug absorption: Inducing emesis, Gastric lavage, Cholestyramine, Chemical inactivation (Giving heparin (-ve) for protamine (+ve)) or Purgative agent.
 - ~ Increasing excretion or elimination of the poison
 - By increasing drug metabolism (biotransformation) --> giving enzyme inducers (eg, ethanol administration for methanol intoxication)
 - Urinary excretion --- if the poison is weak acid, giving alkaline drugs to decrease reabsorption and facilitate excretion
 - Dialysis --- hemodialysis (blood dialysis)
 - Antidote ---- antagonism/chemical inactivation

Causality assessment is done --- if the ADR is associated with a drug.

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